Green subsidies with demand distortions

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Abstract

Demand distortions such as credit constraints and behavioral biases are pervasive in lower-income contexts. A model of clean technology adoption shows that distortions can lower abatement but increase the welfare gains from green subsidies, through two channels: shifting the marginal adopter toward higher private and social benefits, and increasing demand elasticity. We test this by cross-randomizing fixed-cost subsidies, marginal-cost subsidies, and loan access for a US$82 induction stove among 2,134 charcoal users in Kenya. Marginal-cost subsidies of 75% have a precise zero effect on adoption and usage. Credit constraints, however, increase the observable welfare gain per fixed-cost subsidy dollar from US$7.2 to US$12.6 and lower the subsidy cost per tCO_2e from US$19 to US$11. This happens through the two hypothesized channels: increasing the marginal adopter's positive externality by 19% and fuel savings by 30%, and lowering the subsidy cost per marginal adoption by 30%. Estimating the model, we simulate an undistorted counterfactual: each subsidy dollar generates 19 times less observable welfare gain, and the subsidy cost per tCO_2e rises to US$162. Climate financing could be most effective in contexts with larger demand distortions.

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1 Introduction

Many climate change mitigation technologies generate social benefits only if an initial investment is followed by a stream of follow-through decisions: an electric vehicle abates emissions only when driven in place of a gasoline car. When follow-through generates private benefits, as with most energy-saving technologies, the market selects well: the agents who generate the largest social benefits also have the highest willingness-to-pay (WTP) and adopt first.[1] However, this same force creates an adverse selection problem for subsidies. Only agents with low private benefits would choose not to adopt, so adopters who are marginal to the subsidy have lower usage and generate less social benefit. Much of the green subsidy expenditure accrues to inframarginal agents, who would have adopted anyway. These dynamics have dampened the social benefits of a wide range of programs subsidizing upfront investment.[2] A marginal-cost subsidy aims to avoid these problems: by aligning private incentives with social benefits, it draws in adopters in proportion to the abatement they generate and encourages follow-through on the intensive margin (Pigou, 1920).

This logic assumes that WTP reflects private benefit. However, in many settings demand distortions drive WTP below the private benefit (Atkin et al., 2025). Behavioral biases such as present bias, loss aversion, and inattention cause agents to undervalue benefits (Laibson, 1997; DellaVigna, 2009; Duflo et al., 2011). Capital market failures prevent agents from accessing future payment streams to fund present-day investments (de Mel et al., 2008; Banerjee and Duflo, 2014). As a result, even agents with large private benefits from follow-through may not make the initial investment, leaving them marginal to a subsidy. Demand distortions can thus reverse the adverse selection problem, making green subsidies doubly effective: relative to an undistorted market, each dollar can induce more adoptions, and each induced adoption can generate larger social benefits.

To formalize this logic, we first develop a simple model of how demand distortions affect the optimal use of marginal-cost subsidies (which operate on the follow-through margin) and fixed-cost subsidies (which operate on the investment margin) for incentivizing clean technology adoption. The model yields two results: climate financing generates larger welfare gains when targeted towards contexts with larger demand distortions, and fixed-cost subsidies can dominate marginal-cost subsidies in such contexts. We show that, under certain conditions, both hold even if the distortion adds real economic costs.

We evaluate these dynamics empirically in the context of induction cookstove adoption among 2,134 charcoal-cooking households in Nakuru, Kenya. This setting exhibits severe capital market distortions: even with sophisticated credit technologies, buyers paying in installments are normally charged an annualized percentage rate (APR) of 224%. This context allows us to estimate the impact and optimal design of green subsidies by (i) identifying the private and social benefits of a green technology, (ii) varying fixed and marginal-cost subsidies, and (iii) manipulating demand

[1] When follow-through is privately costly, as with forest conservation or sustainable agricultural practices, subsidizing the investment attracts the agents least willing to follow through, and contracting directly on follow-through requires costly information and enforcement capacity and may introduce moral hazard (Oliva et al., 2020; Jack et al., 2025; Aspelund and Russo, 2026; Chen et al., 2025).

[2] See for example Boomhower and Davis (2014), Rapson and Muehlegger (2023), and Hahn et al. (2026).

distortions.

We cross-randomize three treatments: fixed-cost subsidies of either 10% or 75% off the US$82 price; marginal-cost subsidies that lower the price of electricity used to operate the stove by 0%, 25%, or 75% for six months (with no change in the cost of electricity for other uses); and randomized access to an interest-free, four-month loan designed to reduce the wedge between WTP and private benefit. We elicit WTP using an incentive-compatible Becker et al. (1964) mechanism. 29% of study participants bought the stove bundle (which included three pieces of compatible cookware), of whom 67% bought it with a loan, with repayment averaging 86%. We convert fuel expenditures to tons of carbon dioxide-equivalent (tCO_2e) using market prices and local deforestation rates.

We first quantify the induction stove's private and social impacts. High-frequency temperature sensors record a reduction in daily charcoal cooking of 27 minutes, and self-reported monthly charcoal spending decreases by US$10. Self-reported monthly electricity spending increases by US$2.3, which aligns with the induction stove's independent measurements (15 kWh per month at US$0.166 per kWh). Aggregate changes in biomass, electricity, and LPG usage save US$8.9 per month and abate 2.5 tCO_2e per year. Historical usage data from the manufacturer suggest a 2.2-year lifespan.

We find that marginal-cost subsidies have no impact. Despite providing detailed information and drawing attention to cooking costs before eliciting WTP, each US$1 in electricity subsidy has a precisely estimated US$0.0 impact on WTP (95% CI: [−US$0.2, US$0.2]), and we can rule out that a 1% usage price decrease increases electric stove usage by more than 0.12%. These results hold with and without access to a loan. An additional randomized SMS experiment designed to draw attention to the subsidies has no impact, consistent with preferences across cooking technologies driving results.

In this context, the optimal subsidy mix uses only fixed-cost subsidies. Each fixed-cost subsidy dollar generates US$3.5 in fuel savings and US$11 in environmental benefit when valuing CO_2e at a social cost of carbon (SCC) of US$120 per ton (EPA, 2023). The subsidy cost per tCO_2e, US$11, is roughly one-tenth of the SCC. The economic cost of abatement is *negative*: fuel savings exceed the stove's production cost. Our preferred estimate of the aggregate observable welfare gain is US$12.6 per subsidy dollar: conservatively assuming the entire distortion represents resource costs lowers this to US$10.3.

The paper's main finding is that demand distortions cause subsidies to generate substantially larger welfare gains. Average WTP in the control group is only 9.3% of the net present value of future discounted fuel savings: when given access to an interest-free loan this increases to 17%. We find that demand distortions in this context increase the observable welfare benefit of fixed-cost subsidies substantially, from US$7.2 to US$12.6 per dollar of government spending. They furthermore decrease the subsidy cost per tCO_2e from US$19 to US$11.

Distortions operate through both channels identified in the theory. First, marginal adopters facing larger credit distortions generate 30% more in private fuel savings (US$175 versus US$228) and abate 19% more CO_2e (4.8 tCO_2e versus 5.7 tCO_2e). Second, credit constraints shift demand elasticity from −0.9 to −2.3, reducing inframarginal expenditure from 38% to 6.5% of total subsidy spending. This reduces the subsidy expenditure per marginal sale by 30%, from US$93 to US$65.

The experiment only partially relaxes distortions: even with loan access, WTP remains well below discounted private benefits. So, how efficient would fixed-cost subsidies be in a setting without distortions, where WTP equals the private benefit? To answer this, we estimate our model using two-step generalized method of moments (GMM) and simulate counterfactuals across a range of distortions. The model is identified using the randomized variation from the experiment. Simulations indicate that fully eliminating distortions up to the efficient WTP would lower the observable welfare benefit of fixed-cost subsidies from US$12.6 to US$0.7 per subsidy dollar (19 times lower), and raise the subsidy cost per tCO_2e from US$11 to US$162 (14 times higher).

Annual carbon mitigation financing exceeds US$1 trillion (Buchner, 2024). Efficient abatement requires allocating each marginal dollar towards the lowest cost abatement technology. Our results suggest that many of the lowest cost portions of the global abatement supply curve could be found in contexts with large market distortions. For example, the U.S. has a sophisticated financial market for consumer goods, with 80% of new cars sold with financing (Experian, 2023). Through the mechanisms identified in this paper, this absence of credit constraints could partly explain why U.S. electric vehicle subsidies cost over US$1,200 in government expenditure per tCO_2e abated (Hahn et al., 2026)—more than 100 times the US$11 per ton in our setting. Even holding the underlying technology fixed, our results indicate that climate mitigation expenditure could generate substantially larger welfare gains in contexts with more distorted markets.

More than 2 billion people lack modern cooking technologies (IEA, 2024). With billions gaining access to electricity in recent decades, widespread electric cooking has become an achievable policy goal (World Bank, 2024). Despite the importance of this problem, causal evidence is scarce. Using experimental variation, Desbureaux et al. (2025) and Mahadevan et al. (2025) find that fixed-cost subsidies increase adoption of electric cooking technologies in the DRC and in Nepal, respectively, which increases electricity usage and generates important social benefits. Gould et al. (2023) find that the nationwide transition from gas towards electric cooking reduced Ecuador's CO_2e emissions. Other descriptive research and laboratory experiments have shed light on perceptions, usage patterns, energy efficiency, and potential for scale of both induction stoves and pressure cookers (Aemro et al., 2021; Leary et al., 2021; Saha et al., 2021; Clements et al., 2020; Alem et al., 2014; Banerjee et al., 2016; Rubinstein et al., 2022).

While Pigouvian taxes are theoretically first-best (Pigou, 1920), this paper primarily focuses on green subsidies for several reasons. Carbon taxes are often politically intractable: in Canada, a well-designed carbon tax was recently repealed due to political pressures (Department of Finance Canada, 2025). Energy expenditures tend to be inelastic, and low-income households spend a disproportionate share of their incomes on energy costs: in June 2026, Kenya froze electricity tariffs amid "public discontent over fuel prices" (Daily Nation, 2026). Taxation may also be logistically infeasible given the widespread informality of firewood and charcoal production. In addition, many environmental principals (e.g. multilateral donors, environmental agencies, the UNFCCC, carbon offset markets, philanthropists, and investors) lack the legal authority to levy a tax. Their focus is subsidizing climate change mitigation. Doing so efficiently requires identifying technologies and contexts with the lowest marginal abatement costs, at the bottom of the abatement supply curve, and then moving up over time (Gillingham and Stock, 2018; Kolstad, 2011).

Related literature This paper contributes to several literatures. First, we contribute to a growing literature documenting how demand distortions or wedges affect market outcomes (Atkin et al., 2025; Bergquist et al., 2026; Ghatak and Mookherjee, 2025; Indarte et al., 2026). This relates to a large literature in development economics documenting that credit constraints depress adoption of privately beneficial technologies.[3] We build on this by showing that demand distortions can change the efficiency properties of subsidy instruments through selection and inframarginal expenditure.

Second, we extend the literature on selection and mechanism design for optimal subsidy allocation. This literature emphasizes that the welfare consequences of self-selection hinge on the relationship between agents' private willingness to participate and the social value of their actions (Aspelund and Russo, 2026; Chen et al., 2025; Ito et al., 2023). In insurance markets, selection can be adverse or advantageous depending on the correlation between WTP and expected costs (Einav et al., 2010). In carbon-offset and conservation markets, private investment incentives can attract participants whose counterfactual behavior generates little additional abatement, motivating mechanisms that explicitly account for environmental quality and additionality (Mason and Plantinga, 2013). Many green technologies exhibit the opposite pattern: fossil fuel reductions lower private energy expenditures and emissions roughly proportionally, so the same positive correlation that generates advantageous selection into abatement generates adverse selection into subsidy programs. We show that demand distortions mitigate this problem: by driving WTP below private benefit, they reverse the adverse selection facing fixed-cost subsidies.

Third, we contribute to a longstanding literature on optimal environmental policy instruments and second-best environmental policies in the presence of two or more market failures (Jacobsen et al., 2020; Knittel and Sandler, 2018; Holland et al., 2016; Fowlie et al., 2016; Allcott and Taubinsky, 2015). While some papers have shown the benefits of fixed-cost subsidies over marginal-cost subsidies in some contexts (e.g. Carlton and Loury, 1980; De Groote and Verboven, 2019), we show that demand distortions such as behavioral biases and credit constraints can *increase* the efficiency of some green subsidies.

Finally, a large empirical energy economics literature, primarily focused on high-income countries, studies demand elasticities both on the intensive margin (Ito and Zhang, 2025; Ito, 2014; Kahn and Wolak, 2013; Ida et al., 2026; Hahn and Metcalfe, 2021) and on the extensive margin (Houde and Aldy, 2017; Boomhower and Davis, 2014; Borenstein and Davis, 2025). Inattention to energy costs has been documented in the context of cars (Gerarden et al., 2017 and Rapson and Mueh-

[3]Credit constraints have been shown to dampen technology adoption by firms (Banerjee and Duflo, 2014; de Mel et al., 2008; Fafchamps et al., 2014; McKenzie and Woodruff, 2008; Quinn and Woodruff, 2019) and households (Duflo et al., 2008; Anagol and Udry, 2006; Gertler et al., 2024), and in the adoption of green technology in particular (Adhvaryu et al., 2020; Berkouwer and Dean, 2022; Rom et al., 2023; Fowlie and Meeks, 2021). Agents may also face psychological borrowing frictions (Martínez-Marquina and Shi, 2024; Prelec and Loewenstein, 1998; Sussman and Shafir, 2011).

legger, 2023 present overviews), housing (Myers, 2019; Myers et al., 2022), and other household technologies (e.g. Hausman, 1979; Allcott and Taubinsky, 2015; Gillingham and Palmer, 2014). We estimate both margins in a very different context and find a substantially more elastic extensive margin while marginal-cost subsidies move neither WTP nor stove usage, advancing the policy debate around electricity tariffs designed to spur decarbonization. Relatedly, our research design circumvents a technological constraint of subsidies inadvertently incentivizing the use of electricity for other purposes (Borenstein, 2025; Campbell, 2025).

2 Theory

Agents are heterogeneous in their perfectly inelastic demand for energy services θ_i (e.g., transportation or cooking). Agents can meet this demand with an incumbent dirty technology (e.g., a gasoline vehicle or a charcoal cookstove), or buy a cleaner technology (e.g., an electric vehicle or an electric cookstove) and then shift some fraction of their energy consumption to the clean technology. Per unit of energy service, the dirty technology incurs fuel costs f_H and emits ϕ_H. The clean technology costs $f_L < f_H$ and emits $\phi_L < \phi_H$, and can be bought at marginal cost p. Define, respectively, the per-unit cost saving and the per-unit externality reduction from using the clean technology:

$$\pi \equiv f_H - f_L > 0 \qquad\qquad \phi \equiv \phi_H - \phi_L > 0$$

Throughout, emissions are valued at the social cost of carbon, and ϕ denotes the dollar value of the per-unit externality reduction.

We start by presenting graphical intuition, and then formalize the theory to establish three results. First, for any given subsidy level, demand distortions raise the abatement and welfare generated per subsidy dollar (Section 2.3). Second, distortions tilt the optimal instrument mix toward fixed-cost subsidies (Section 2.4). Third, under some conditions, these results hold even when the wedge generates a real resource cost (Section 2.5).

2.1 Graphical intuition

We first present a pared-down version of the model in which adopters use the green technology to meet all their energy services demand. A marginal-cost subsidy s_2 raises the per-unit saving from π to $\pi + s_2$, so willingness-to-pay (WTP) is the private marginal benefit $wtp_i = (\pi + s_2)\theta_i$, and market demand is $Q(p) = \Pr(wtp_i > p)$. A fixed-cost subsidy s_1 induces adoption by marginal adopters with $p - s_1 < wtp_i < p$.

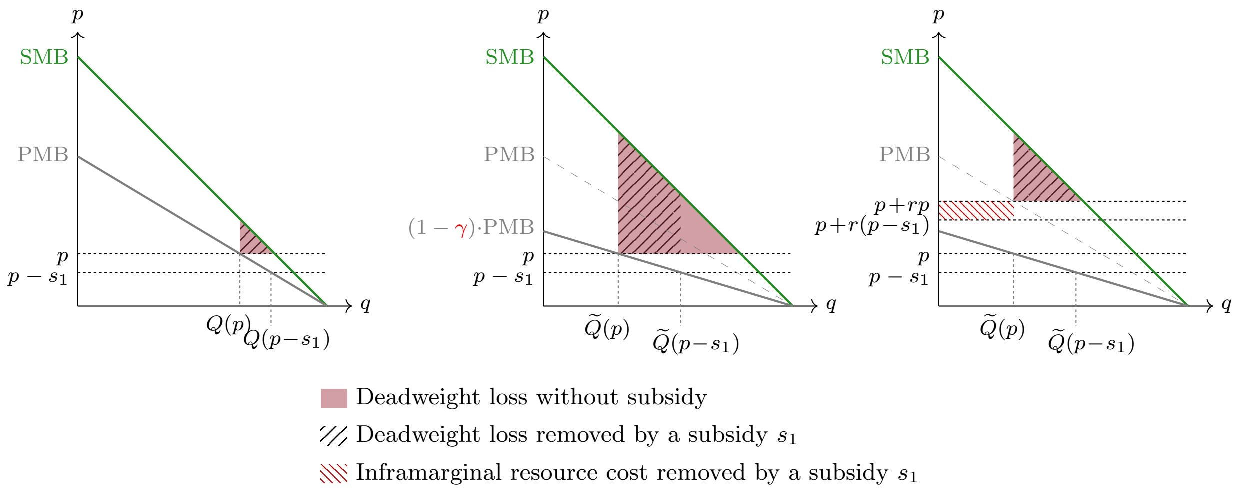
We consider two types of distortion, which generate identical behavior but have different welfare implications (Figure 1). An allocation distortion ρ changes decision-making but not payoffs: examples include inattention to benefits and present bias. A resource distortion r consumes resources: each dollar the agent spends on the technology requires society to expend a further r dollars. Write

Figure 1: How demand distortions affect deadweight loss at equilibrium and with a subsidy

(a) Without distortion ($\gamma = 0$)

(b) Allocation distortion ($\rho > 0$, $r = 0$)

(c) Resource distortion ($\rho = 0$, $r > 0$)



Notes: Distortions increase both the deadweight loss (DWL) at equilibrium and the DWL that a subsidy removes. Panels (b) and (c) share a demand curve and adoption cutoffs, and differ only in which piece of γ is positive. Panel (c) has two benchmarks because s_1 induces adoption and also lowers the resource cost. The striped maroon rectangle has height $r\,s_1$ and represents the total resource cost reduction for inframarginal adopters.

the adoption rule with one wedge on each side:

$$(1-\rho)\,wtp_i \;>\; (1+r)\,(p-s_1)$$

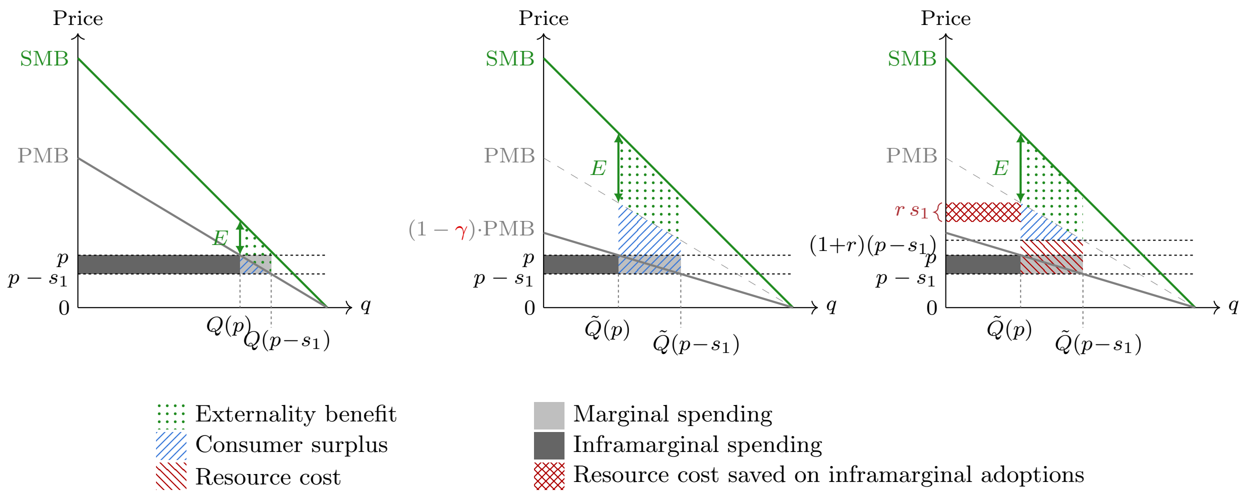
The pure cases are the corners $r = 0$ and $\rho = 0$. Their joint impact is summarized by the composite distortion $\gamma \equiv (\rho + r)/(1 + r) \in [0, 1)$, which scales down WTP at the time of purchase: $\widetilde{wtp}_i = (1-\gamma)\,wtp_i$. Under an allocation distortion (panel b) the entire gap between PMB and $(1-\gamma)$·PMB is unrealized surplus. Under a resource distortion (panel c) most of that gap is a resource cost, and the remaining loss is only the uninternalized externality.

We explore two channels through which distortions might increase subsidy efficiency: changing the composition of marginal adopters and changing demand elasticity.

Channel 1: The marginal adopter generates a larger fuel use reduction Define the marginal adopter as $\underline{\theta}\ :\ (1-\gamma)(\pi + s_2)\,\underline{\theta} = p - s_1$. An increase in the distortion pushes some high-usage agents below the adoption threshold, increasing the marginal adopter's usage. Since abatement is proportional to usage ($\phi_i = \theta_i\phi$), this increases the positive externality generated by the marginal adopter. Figure 2 shows this: whenever WTP and abatement are positively correlated, the marginal adopter's externality (E) is larger under the distortion (Section 2.8 discusses the negative case).

Figure 2: Effect of demand distortions on the welfare gain from a fixed-cost subsidy

(a) Without distortion ($\gamma = 0$) (b) Allocation distortion ($\rho > 0$, $r = 0$) (c) Resource distortion ($\rho = 0$, $r > 0$)



Notes: Private Marginal Benefit (PMB), Social Marginal Benefit (SMB), and the externality (E) generated by the marginal adopter. The dotted green area is the total externality generated by induced adopters, the striped blue area their consumer surplus, and the gray areas the expenditure when subsidizing the price to $p - s_1$. Panel (a) is the case without distortions. Panel (b) shows a pure allocation distortion ($r = 0$). Panel (c) shows a pure resource distortion ($\rho = 0$): the red band is resources consumed, the triangle above it is surplus, and the crosshatched rectangle is the resource cost $r\,s_1$ saved on each inframarginal adoption. E is identical in (b) and (c). Figure A1 shows selection under negatively correlated private and social benefits.

Channel 2: Demand distortions increase demand elasticity Demand distortions can increase demand elasticity, which decreases the inframarginal share of subsidy recipients and reduces the cost per marginal adopter. Figure 2 illustrates this intuition graphically for the fixed-cost subsidy: the distortion shrinks the inframarginal region (dark) relative to the marginal region (light). The distorted elasticity at price p equals the undistorted elasticity at the higher price $p/(1-\gamma)$: $\tilde{\mathcal{E}}(p) = \mathcal{E}(p/(1-\gamma))$. Because $|\mathcal{E}|$ is increasing in price under an increasing hazard rate, distortions increase demand elasticity.[4]

2.2 Formal set-up

Policy instruments We consider three policy instruments: (1) a fixed-cost subsidy s_1, reducing the one-time adoption price to $p - s_1$; (2) a marginal-cost subsidy s_2 on the cleaner fuel; (3) a marginal cost tax τ on the dirtier fuel. Define the relative fuel price signal facing the agent:

$$\pi' = (f_H + \tau) - (f_L - s_2) = \pi + \tau + s_2$$

[4] A corner solution is possible here where $Pr((1-\gamma)wtp > p) = 0$ such that there is no demand at the competitive price. Any subsidy s_1 with $Pr((1-\gamma)wtp > p - s_1) = 0$ would then trivially have no impact on adoption since elasticity is undefined at $q = 0$. The model assumes an interior solution, which reflects the data as well as the fact that any corner solution is an artefact of the assumed linear demand curve and homogenous distortion.

The agent's problem Agents are distributed $g(\alpha)$ by a (dis)taste parameter α_i for the clean technology.[5] Utility is quasi-linear with no discounting.

At $t = 1$, agent i decides whether to buy the clean technology at $p - s_1$: $d_{1i} \in \{0, 1\}$. If they adopt, at $t = 2$ they choose the share of energy services to shift to the clean technology: $d_{2i} \in [0, 1]$. Each additional unit shifted to the clean technology generates marginal non-pecuniary switching cost $\mu'(d_{2i}\theta_i - \alpha_i)$, such as learning or habit change, where $d_{2i}\theta_i - \alpha_i$ denotes clean energy services consumed in excess of their taste parameter α_i. Higher-α_i agents have less distaste for the clean technology. We assume $\mu(0) = 0$, $\mu'(0) = 0$, $\mu'' < 0$, and $\lim_{x\to\infty} \mu'(x) = -\infty$, so that $\mu(x) \leq 0$ for all $x \geq 0$, and that switching costs are sufficiently curved, $|\, x\,\mu''(x)/\mu'(x)\,| > 1$ for all $x > 0$.[6] Total utility from usage conditional on adoption is:

$$f(d_{2i}, \pi', \theta_i, \alpha_i) = d_{2i}\theta_i\pi' + \mu(d_{2i}\theta_i - \alpha_i)$$

The agent shifts clean energy consumption until marginal disutility equals the per-unit fuel saving: $\mu'(d^*_{2i}\theta_i - \alpha_i) = -\pi'$. Since π' and $\mu(\cdot)$ are common across agents, so is usage in excess of taste; denote it x^*. We restrict attention to allocations in which the marginal adopter has nonnegative taste, and assume $x^* + \alpha_i < \theta_i$ throughout the support, so that every adopter's usage is at least x^* and interior ($d^*_{2i} \in (0, 1)$). Define willingness-to-pay (WTP) as follows:

$$wtp_i = (x^* + \alpha_i)\pi' + \mu(x^*) = x^*\pi' + \mu(x^*) + \alpha_i\pi' \tag{1}$$

The agent adopts iff $wtp_i > p - s_1$. Market demand is $\Pr(wtp_i > p - s_1)$.

We assume α is distributed with an increasing hazard rate (IHR) such that demand is log-concave.[7] Agents with a stronger taste use the technology more, but WTP does not depend on θ_i because heavy users substitute up to the same bliss point.

Lemma 1. *Willingness-to-pay is increasing in α_i and does not depend on θ_i: heterogeneity in both usage and willingness-to-pay depends only on α_i.* *Proof in Appendix VI.1.*

Demand distortions The composite distortion γ scales down WTP but leaves usage conditional on adoption unchanged:

$$\widetilde{wtp}_i = (1 - \gamma)\, wtp_i = (1 - \gamma)[(x^* + \alpha_i)\pi' + \mu(x^*)]$$

Define the marginal adopter as $\underset{\sim}{\alpha} \,:\, \widetilde{wtp}_i(\underset{\sim}{\alpha}) = p - s_1$. A fixed-cost subsidy lowers the adoption cutoff from $\underset{\sim}{\alpha}_0$, the cutoff at the unsubsidized price p, to $\underset{\sim}{\alpha}$, the cutoff at $p - s_1$. The $\Delta Q \equiv G(\underset{\sim}{\alpha}_0) - G(\underset{\sim}{\alpha})$ agents in between are induced by the subsidy to adopt, while s_1 is paid to all

[5] This framework builds on Allcott, Mullainathan, and Taubinsky (2012; 2014).

[6] Because the agent's optimal usage equates $\mu'(x^*) = -\pi'$, this requires the elasticity of x^* with respect to the reward to be below one. Estimated elasticities are approximately 0.1–0.3 (Allcott et al., 2012), well below one.

[7] This assumes demand elasticity monotonically increases in price, a relatively common assumption that rules out heavy-tailed usage distributions. It is satisfied for example by the uniform, normal, and exponential distributions.

$Q \equiv 1 - G(\underline{\alpha})$ buyers at the subsidized price. Two statistics summarize the subsidy: the average usage of induced adopters,

$$\bar{u} \;\equiv\; \mathbb{E}\big[x^* + \alpha \;\big|\; \alpha \in [\underline{\alpha}, \underline{\alpha}_0]\big],$$

equivalently $\bar{u} = x^* + \bar{\alpha}$, where the mean taste of induced adopters is $\bar{\alpha} \equiv \mathbb{E}[\alpha \mid \alpha \in [\underline{\alpha}, \underline{\alpha}_0]]$; and the number of recipients per induced adoption, $\zeta \equiv Q/\Delta Q \geq 1$. Fixed-cost spending per induced adoption is then $s_1\zeta$, of which $s_1(\zeta - 1)$ reaches inframarginal recipients. Its counterpart at the margin is $\kappa \equiv Q/(\partial Q/\partial s_1)$, the inframarginal spending per induced adoption at the margin.

The principals Define total clean energy usage $U \equiv \int_{\underline{\alpha}}^{\infty}(x^* + \alpha_i)\, g(\alpha)\, d\alpha$ and total subsidy expenditure $C \equiv s_1 Q + s_2 U$; both depend on $\{s_1, s_2\}$ through $\underline{\alpha}$ (who adopts) and x^* (how much each adopter uses). A constrained principal has no taxation authority and a budget constraint $C \leq B$, choosing $\{s_1, s_2\}$ to maximize the value generated by adoption:

$$\{s_1^*, s_2^*\} = \arg\max \int_{\underline{\alpha}}^{\infty} \mathcal{V}(\alpha_i)\; g(\alpha)\, d\alpha$$

We consider two principals. The environmental principal values only abatement, $\mathcal{V} \equiv (x^* + \alpha_i)\,\phi$, so their aggregate value reduces to ϕU. A social principal values the full welfare from adoption:

$$\mathcal{V}(\alpha_i) \;\equiv\; \underbrace{(x^* + \alpha_i)\phi}_{\substack{\text{externality}\\ \text{reduction}}} \;+\; \underbrace{(x^* + \alpha_i)\pi + \mu(x^*)}_{\substack{\text{private value}\\ \text{(true price, actual usage)}}} \;-\; \underbrace{p}_{\substack{\text{technology}\\ \text{cost}}}$$

The unconstrained social planner has this same value function but can both subsidize and tax.

2.3 Distortions raise the return per subsidy dollar

We first hold the instruments $\{s_1, s_2\}$ constant and ask how the distortion moves the return per dollar of subsidy expenditure. Distortions lower welfare levels (the deadweight loss in Figure 1) but the object relevant for allocating subsidy dollars is the marginal return per subsidy dollar.

Lemma 2. *For a given set of instruments, demand distortions raise the adoption cutoff $\underline{\alpha}$ and hence the marginal adopter's usage:*

$$\frac{\partial(x^* + \underline{\alpha})}{\partial\gamma} > 0$$

Proof in Appendix VI.2.

Lemma 3. *For a given set of instruments, the distorted demand elasticity at any price P equals the undistorted elasticity at the scaled-up price, so under the maintained IHR assumption demand distortions raise demand elasticity and lower the inframarginal cost per induced adoption:*

$$\tilde{\mathcal{E}}(P) = \mathcal{E}\left(\frac{P}{1-\gamma}\right), \qquad \frac{\partial\,|\tilde{\mathcal{E}}(P)|}{\partial\gamma} > 0, \qquad \frac{\partial\kappa}{\partial\gamma} < 0$$

Proof in Appendix VI.1.

Write Δ for a change in s_i relative to $s_i = 0$, holding s_j constant. We assume a purely allocative distortion ($r = 0$) here and in Section 2.4 (Section 2.5 relaxes this).

Fixed-cost subsidy Expenditure C increases through s_1 directly plus s_2 paid on new adopters. Both principals' objectives are functions of usage, so we state the results in usage per subsidy dollar.

Proposition 1. *For a given set of instruments, larger demand distortions raise both the average and the marginal usage generated per dollar of fixed-cost subsidy expenditure:*

(i) average usage per dollar,

$$\frac{\Delta U}{\Delta C} = \frac{\bar{u}}{s_1\zeta + s_2\bar{u}},$$

is strictly increasing in γ;

(ii) marginal usage per dollar,

$$\frac{dU/ds_1}{dC/ds_1} = \frac{x^* + \underline{\alpha}}{s_1 + s_2(x^* + \underline{\alpha}) + \kappa}, \qquad \kappa = \frac{(1-\gamma)\pi'[1 - G(\underline{\alpha})]}{g(\underline{\alpha})} = \frac{p - s_1}{|\tilde{\mathcal{E}}(p - s_1)|},$$

is strictly increasing in γ at any interior margin. *Proof in Appendix VI.2.*

Distortions raise $\bar{u}$ (selection) and lower ζ (elasticity), so the average return (i) rises (Figure 2). In (ii), the marginal adopter's usage rises (Lemma 2) and $|\tilde{\mathcal{E}}|$ rises (Lemma 3), so inframarginal leakage κ falls. The average welfare gain per fixed-cost subsidy dollar is also increasing in the allocation distortion ρ, provided the average induced adoption has positive social value (Section VI.4).

Marginal-cost subsidy The marginal-cost subsidy operates on both margins. When $s_2 = 0$, the cutoff is $\underline{\alpha}^\circ$ and usage is $x^\circ < x^*$. Define the s_2-margin counterparts of $\bar{u}$ and ζ: $\bar{u}_2 \equiv \mathbb{E}[x^* + \alpha \mid \alpha \in [\underline{\alpha}, \underline{\alpha}^\circ]]$ and $\zeta_2 \equiv Q/(Q - Q^\circ)$ with $Q^\circ = 1 - G(\underline{\alpha}^\circ)$, and mean usage among all adopters $\bar{u}_Q \equiv U/Q$.

Proposition 2. *For a given set of instruments, larger demand distortions raise both the average and the marginal usage generated per dollar of marginal-cost subsidy expenditure:*

(i) average usage per dollar,

$$\frac{\Delta U}{\Delta C} = \frac{\bar{u}_2 + (\zeta_2 - 1)(x^* - x^\circ)}{s_1 + s_2\,\zeta_2\,\bar{u}_Q},$$

is strictly increasing in γ, provided the usage-weighted density $(x^\circ + \alpha)\,g(\alpha)$ inherits the IHR, and $x^\circ + \underline{\alpha}^\circ \geq \bar{u}_2 + (\zeta_2 - 1)(x^* - x^\circ)$: the marginal adopter when $s_2 = 0$ uses at least the usage generated per induced adoption;

(ii) marginal usage per dollar,

$$\frac{dU/ds_2}{dC/ds_2} = \frac{\frac{dU}{ds_2}}{U + s_2\frac{dU}{ds_2} + s_1\,g(\underline{\alpha})\left(-\frac{\partial\underline{\alpha}}{\partial s_2}\right)}$$

is strictly increasing in γ at any interior margin, provided the fixed-cost subsidy does not exceed the marginal adopter's willingness-to-pay. *Proof in Appendix VI.2.*

One might expect a distortion to lower the return to the marginal-cost subsidy, since a dollar of s_2 raises the marginal adopter's effective WTP by only $(1-\gamma)(x^*+\underline{\alpha})$. But the distortion scales down all components of WTP equally, by the same factor $(1-\gamma)$ across agents, making demand more price-responsive by exactly $1/(1-\gamma)$ (the elasticity channel). A dollar of s_2 buys a smaller effective price cut, but each unit of price cut buys proportionally more adoptions, and the two cancel exactly. The distortion has no direct effect on adoptions per dollar of s_2, $g(\underline{\alpha})(x^*+\underline{\alpha})/\pi'$; γ enters only through the cutoff $\underline{\alpha}$ (Figure A2). The intensive margin is also unaffected, since the distortion does not enter the usage decision.

What changes is the adopter pool, through the same two channels as for the fixed-cost subsidy. Selection: a deeper distortion raises the cutoff, so each adoption the subsidy induces generates more usage. Elasticity: adoption falls while the marginal responses do not, shrinking the inframarginal usage U on which most s_2 expenditure is paid. Both channels raise usage per dollar, so the distortion raises the return to s_2.

A counteracting force works in the opposite direction: the remaining adopters are heavier users, which raises the s_2 bill per adopter, and in (i) the intensive-margin gain (x^*-x°) is collected on fewer inframarginal adoptions as ζ_2 falls. The conditions in the proposition bound these terms, so they do not reverse the effect.

2.4 Distortions tilt the optimal instrument mix toward fixed-cost subsidies

We now let the principals choose the mix $\{s_1, s_2\}$ under the budget, again maintaining a purely allocative distortion (relaxed in Section 2.5). A dollar of s_1 lowers the price the agent compares against perceived WTP one-for-one, whereas a dollar of s_2 raises the marginal adopter's perceived WTP by only $(1-\gamma)(x^*+\underline{\alpha})$. However, only s_2 generates value on the intensive margin by raising usage. At an interior optimum the principal equates the two instruments' marginal value per dollar:

$$\frac{\overbrace{\mathcal{V}(\underline{\alpha})}^{\substack{\text{value per}\\ \text{new adopter}}}}{\underbrace{s_1+s_2(x^*+\underline{\alpha})}_{\substack{\text{total subsidy to}\\ \text{new adopters}}}+\underbrace{\kappa}_{\substack{\text{inframarginal}\\ \text{cost}}}} = \frac{\overbrace{\frac{\partial \mathcal{V}}{\partial s_2}}^{\substack{\text{value}\\ \text{of additional usage}}}Q \quad + \overbrace{\mathcal{V}(\underline{\alpha})g(\underline{\alpha})\left(-\frac{\partial \underline{\alpha}}{\partial s_2}\right)}^{\substack{\text{value}\\ \text{of new adopters}}}}{\underbrace{s_2\frac{dx^*}{ds_2}Q}_{\substack{\text{cost of}\\ \text{increased usage}}} + \underbrace{\left[s_1+s_2(x^*+\underline{\alpha})\right]g(\underline{\alpha})\left(-\frac{\partial \underline{\alpha}}{\partial s_2}\right)}_{\substack{\text{total subsidy to}\\ \text{new adopters}}} + \underbrace{U}_{\substack{\text{inframarginal}\\ \text{cost}}}} \tag{2}$$

Derivation in Appendix VI.4.

The left-hand side is the marginal adopter's value per dollar of fixed-cost subsidy spent inducing them: a dollar of s_1 buys $g(\underline{\alpha})/((1-\gamma)\pi')$ adoptions—the full responsiveness gain with no damped price cut—and distortions further raise this side by increasing the marginal adopter's value $\mathcal{V}(\underline{\alpha})$ and lowering inframarginal leakage κ. The right-hand side is the marginal-cost margin: distortions raise this less, because they have no direct effect on the adoptions per dollar of s_2; the damping of the price signal and the compression of perceived WTP offset exactly. Both movements shift the optimal mix toward the fixed-cost subsidy.

The binding budget leaves one degree of freedom: choosing s_2 pins down s_1 and the cutoff $\underline{\alpha}$. A distortion raises s_1^* through two channels: holding s_2 constant, maintaining adoption at the same cutoff requires a higher s_1; and the principal lowers s_2^*, freeing budget for s_1.

Proposition 3. *At an interior optimum satisfying a strict second-order condition, a deeper allocation distortion increases the social principal's optimal fixed-cost subsidy s_1^*:*

$$\left.\frac{ds_1^*}{d\gamma}\right|_{r=0} = \underbrace{\left.\frac{\partial s_1}{\partial \gamma}\right|_{s_2}}_{\textit{direct effect}} + \underbrace{\left.\frac{\partial s_1}{\partial s_2}\right|_{\gamma} \cdot \frac{ds_2^*}{d\gamma}}_{\textit{re-optimization}} > 0$$

Proof in Appendix VI.4.

The aggregate efficacy of a re-optimized subsidy pair is governed by the difference between aggregate usage under the subsidy and under a no-subsidy counterfactual. By the envelope theorem, every re-optimization term collapses into the budget's shadow value λ (the marginal usage per budget dollar). The aggregate effect of a distortion thus depends on whether it increases emissions faster in the no-subsidy scenario or in the optimized subsidy scenario.

Proposition 4. *Let $V(\gamma)$ denote maximized usage under the budget B and $U^\circ(\gamma)$ usage in the no-subsidy counterfactual. At an optimum interior in s_1:*

(i) Every re-optimization term collapses into the budget's shadow value λ, and the distortion reduces maximized usage at rate

$$\frac{dV}{d\gamma} = -\lambda\, wtp(\underline{\alpha})\, Q.$$

(ii) Usage generated per budget dollar, $[V(\gamma) - U^\circ(\gamma)]/B$, is strictly increasing in γ if and only if the counterfactual's loss rate exceeds $\lambda\, wtp(\underline{\alpha})\, Q$. This holds under a bound on the decline of the taste density between the subsidized and the no-subsidy margins, and in particular at all sufficiently small instrument levels.

Proof in Appendix VI.5.

Three forces push the usage generated per budget dollar to rise with γ. First, each adoption removed by the distortion reduces abatement by the exiting adopter's usage, but also reduces the principal's spending by $s_1 + s_2(x^* + \underline{\alpha})$: they can reallocate this at marginal value λ, so the principal loses only the inframarginal leakage $\lambda\kappa$ per exit. Second, the distortion scales perceived WTP by $(1-\gamma)$ at both margins (as in Figure A2), but the subsidy steepens the profile of willingness-to-pay across agents—it rises at rate π' rather than π—so a given distortion expels fewer adopters at the subsidized margin than at the no-subsidy margin. Third, the marginal adopter at the no-subsidy margin has both higher WTP (so the distortion removes adopters there at a faster rate) and generates higher benefit (so each exit reduces more welfare).

Under the additional assumption that enough agents remain near the no-subsidy margin (Appendix VI.5), the counterfactual loss dominates: the return to subsidy spending increases with γ

even when the principal re-optimizes the instrument mix. A reversal requires the density at the subsidized margin to exceed the density at the no-subsidy margin by more than the product of the three attenuation factors.

2.5 Evaluating these results with a resource distortion

Now let $r > 0$: every adoption consumes real resources. Conditional on holding γ fixed, r does not affect the agent's decision, and the environmental principal places no weight on it. Every environmental result holds verbatim at any r: Lemma 2, Propositions 1 and 2, and the environmental principal's instrument-mix comparative static (Appendix VI.3).

For the social principal the resource cost enters twice. On the one hand it lowers the value of each induced adoption: $\mathcal{V}^R(\alpha_i; s_1) \equiv \mathcal{V}(\alpha_i) - r\,(p - s_1)$. On the other, a larger subsidy lowers the resource cost of every inframarginal adoption ($\partial\mathcal{V}^R/\partial s_1 = r$). The welfare generated per subsidy dollar is then a function of the average value generated by induced adopters, $\mathcal{V}^R(\bar{\alpha})$:

$$\frac{\Delta W}{\Delta C} = \frac{\overbrace{\mathcal{V}^R(\bar{\alpha})}^{\text{value of induced adoptions}} + \overbrace{s_1(\zeta - 1)\,\frac{\partial \mathcal{V}^R}{\partial s_1}}^{\text{resource cost saved on inframarginal adoptions}}}{s_1\zeta + s_2\bar{u}},$$

A larger resource distortion tends to increase the welfare gain per subsidy dollar when the subsidy covers most of the price at the margin, when induced adoptions are valuable well beyond their additional resource cost, and when the marginal adopter's usage rises quickly with the distortion. Conversely, it falls when the inframarginal base is large and the taste distribution thins out so that the targeting gains stall.

Proposition 5. *Hold the instruments fixed and let $r \geq 0$. Assuming the technology cost exceeds the fixed-cost subsidy paid per induced adoption ($s_1\zeta \leq p$), and the average induced adoption's value covers r times the average total subsidy ($\mathcal{V}^R(\bar{\alpha}) \geq r(s_1 + s_2\bar{u})$),*

(i) The welfare gain per subsidy dollar is increasing in the allocation distortion ρ at any fixed r.

(ii) Deepening the resource distortion r itself adds a direct cost term absent from the ρ-experiment; welfare per subsidy dollar remains increasing only under an additional bound.

Proof in Appendix VI.6.

The quantity $s_1\zeta - p$ is the net effect of the resource distortion on welfare per subsidy dollar when factoring in the resources saved by shrinking every recipient's payment against the resource cost of the adoptions the subsidy induces. The proposition assumes that direct effect is a cost, which is why part (ii) needs a further bound. When leakage is heavy enough that the inequality runs the other way, the direct effect reinforces the selection and elasticity channels instead and no further bound is required—the case treated next.

There are two pointwise conditions under which a resource distortion $r > 0$ unambiguously increases the efficacy of a fixed-cost subsidy s_1 with $s_2 = 0$. First, the average induced adoption's value covers r times the fixed-cost subsidy. Second, leakage must be large enough that the program spends at least the full technology price in fixed-cost subsidies per adoption it induces.

Corollary 1. *Suppose $s_2 = 0$. If, at the prevailing equilibrium, the average induced adoption's value covers r times the fixed-cost subsidy ($\mathcal{V}^R(\bar{\alpha}) \geq rs_1$) and the program spends at least the technology cost in fixed-cost subsidies per adoption it induces ($s_1\zeta \geq p$), then welfare per subsidy dollar is increasing in both the allocation distortion and the resource distortion at that point.*

Proof in Appendix VI.6.

At the margin, the resource distortion raises the return to the fixed-cost subsidy by saving resource costs on the inframarginal adoptions it reaches: each induced adoption adds κr to the fixed-cost side of (2). The tilt carries over to the optimal mix, at least for small distortions: at $r = 0$ a deeper resource distortion strictly raises the optimal fixed-cost subsidy, $ds_1^*/dr > 0$, with the direct and re-optimization channels reinforcing, and the comparative static of Proposition 3 likewise extends to all $r \in [0, \bar{r})$. Both signs are established near $r = 0$ only (Appendix VI.6).

2.6 The social planner

The social planner values adoption at the same $\mathcal{V}^R(\alpha_i; s_1)$ as the social principal but can also tax, choosing $\{s_1, s_2, \tau\}$ to maximize social welfare. First-best usage sets the marginal relative-price wedge equal to the externality: $\tau^* + s_2^* = \phi$ (Pigou, 1920). Under an allocation distortion, $s_1 = p\gamma$ restores the first best. However, under a resource distortion, efficiency requires $wtp_i \geq p + \frac{\gamma}{1-\gamma}(p - s_1)$. The two thresholds differ by s_1: any positive fixed-cost subsidy induces over-adoption, and the marginal adopter destroys s_1 of value. No subsidy restores the first best, because s_1 moves the adoption margin and the resource cost together. The planner still sets $s_1 > 0$, trading over-adoption at the margin against resources saved on every adoption.

Proposition 6. *The optimal marginal tax/subsidy wedge is Pigouvian and unaffected by demand distortions, $\tau^* + s_2^* = \phi$. Both distortion types independently motivate a positive fixed-cost subsidy:*

$$s_1^* = \underbrace{p\,\frac{\rho\,(1+r)}{1+r\rho}}_{\text{corrects the wedge}} + \underbrace{\frac{(1-\rho)^2\,r}{(1+r)\,(1+r\rho)} \cdot \frac{\pi+\phi}{h(\underline{\alpha})}}_{\text{economizes on resource cost}}, \qquad h \equiv \frac{g}{1-G},$$

so that s_1^* collapses to $p\rho$ at $r = 0$ and to $\frac{r}{1+r}\frac{\pi+\phi}{h(\underline{\alpha})}$ at $\rho = 0$. *Proof in Appendix VI.7.*

The allocation term is a corrective transfer: it closes a wedge, costs nothing at the margin, and is pinned by the technology cost p. The resource term is an investment in avoiding a real cost, pinned by the marginal social value of usage $\pi + \phi$ and the inverse hazard rate $1/h(\underline{\alpha})$—the same object that governs inframarginal leakage κ. The first-order condition balances the r saved on each existing adoption against the marginal adopters, each destroying s_1. Applying $p\rho$ to a resource distortion therefore sets the subsidy on the wrong basis and, in general, at the wrong level. s_1^* rises strictly with γ when $r = 0, \rho > 0$ but is more complicated to sign when $r > 0$.[8]

2.7 Spending to reduce the distortion

Should the principal address the distortion directly? We compare three policies against s_1 spending.

Correcting an allocation distortion Let $r = 0$ and suppose the principal can spend I correcting ρ, with $\rho'(I) < 0$ and $\rho''(I) > 0$. Since γ does not enter the usage decision, this is a purely extensive-margin instrument. With $N_\rho \equiv g(\underset{\sim}{\alpha})(p - s_1)|\rho'(I)|/((1-\rho)^2\pi')$ adoptions bought per dollar, the shadow values for correction versus fixed-cost subsidies are

$$\lambda_\rho = \frac{\mathcal{V}(\underset{\sim}{\alpha})}{\underbrace{1/N_\rho}_{\text{cost per induced adoption}} + s_1 + s_2(x^* + \underset{\sim}{\alpha})} \qquad \text{versus} \qquad \lambda = \frac{\mathcal{V}(\underset{\sim}{\alpha})}{\underbrace{\kappa}_{\text{inframarginal leakage}} + s_1 + s_2(x^* + \underset{\sim}{\alpha})}.$$

Derivations in Appendix VI.8.

The numerators coincide, so correction is preferred iff $1/N_\rho < \kappa$, i.e., when conversions are cheap and inframarginal spending per marginal adoption is large.

The only advantage of correcting ρ over the fixed-cost subsidy is the leakage it avoids. Its benefit therefore falls with inframarginal leakage κ (Proposition 1): in very distorted markets most subsidy dollars already reach marginal adopters, and κ falls toward zero. An unconstrained planner never corrects ρ: s_1^* restores the adoption margin with a transfer while correction consumes resources.

Correcting a resource distortion Let $\rho = 0$ and suppose the principal can spend J lowering r, with $r'(J) < 0$ and $r''(J) > 0$, and $N_r \equiv g(\underset{\sim}{\alpha})(p - s_1)|r'(J)|/\pi'$ adoptions bought per dollar. Both policies now save real resources in addition to inducing adopters. A subsidy dollar lowers the price the agent pays, so the resource cost falls at rate r per dollar, saving κr per induced adoption among inframarginal adopters. Lowering r saves $(p - s_1)$ on every adoption per unit removed; per induced adoption this saves $(1+r)\kappa$. At equal cost per conversion ($1/N_r = \kappa$), correcting r therefore dominates the subsidy: it can be worthwhile even when its cost per conversion exceeds κ. For the environmental principal, who does not value the resource savings, the criterion remains $1/N_r < \kappa$.

Subsidizing a resource distortion Alternatively, the principal can subsidize the resource distortion rather than remove it, covering ς of each unit of resource cost, as subsidized credit does. The agent then faces $r - \varsigma$ and adopts iff $wtp_i > (1 + r - \varsigma)(p - s_1)$. A fixed-cost subsidy and a distortion subsidy both reach every buyer, so only the threshold movement per dollar differs: s_1 delivers $(1+r)$ times as much, and the distortion subsidy is dominated, increasingly so as r rises. The advantage

[8] Only the $r = 0$ comparative static is exact ($ds_1^*/d\rho = p > 0$): the resource term is implicit, because $h(\underset{\sim}{\alpha})$ is evaluated at an endogenous cutoff that moves with the distortion. A deeper resource distortion raises the cost to economize on but also raises the cutoff, and when the hazard rises steeply enough the second effect dominates, so that s_1^* falls as r deepens (Appendix VI.7 gives an example at a regular interior optimum with an increasing hazard).

rests on the resource cost being proportional: were it a constant amount per adoption, s_1 would lower the threshold by one dollar and the advantage would vanish.

2.8 Model discussion and possible extensions

Heterogeneous distortions When γ_i varies the two instruments no longer induce the same agents. The most distorted agents have the highest adoption thresholds, so an intervention targeted at them induces high-α_i, hence high-value, adopters, whereas a uniform s_1 moves every agent's threshold equally. A targeting term therefore survives the cancellation and correction performs better than the common-γ criterion suggests. This is Channel 1 operating across agents rather than through the level of γ, and it inherits the same dependence on positive correlation.

Marginal distortions A marginal distortion scales the relative fuel price the agent perceives, to $\delta\pi'$ with $\delta \in (0, 1]$: inattention to fuel costs, or mistaken beliefs about savings. Adopters underuse the technology ($\widetilde{x}^* < x^*$), and WTP falls by more than proportional scaling because the agent discounts the savings but bears the switching costs in full. Both channels persist: the marginal adopter uses more, and demand elasticity rises—though the latter now needs the IHR rather than only $|\mathcal{E}|$ increasing in price. The aggregate result is no longer signed, because a smaller distortion raises usage for every adopter while shifting the induced range down, so selection opposes itself. And unlike γ, a marginal distortion is correctable by price: the planner's optimal wedge exceeds the Pigouvian level ($\tau^* + s_2^* = [\phi + \pi(1-\delta)]/\delta$), restoring both margins at once and leaving $s_1^* = 0$. Correcting γ needs a fixed-cost subsidy precisely because γ also scales switching costs, which no fuel-price instrument affects (Appendix VII).

Non-usage benefits Allowing non-usage benefits to clean technology adoption ($\mu(0) = c$) leaves the basic mechanics intact when clean fuel is privately cheaper, but it does not leave every comparative static unchanged. Usage depends on μ' rather than on the level of $\mu(0)$, so x^* is unchanged, WTP remains affine in α_i, and WTP and abatement both remain increasing in α_i. The social planner's first-best result is likewise unchanged. The form of the instrument-mix condition also carries through, but its components shift with c. However, the proof of the optimal-policy comparative static uses the sign condition $\mu(x^*) < 0$; with sufficiently large non-pecuniary benefits this condition can fail, so that comparative static is stated under this maintained assumption rather than for arbitrary c (Appendix VIII).

Negative correlation between private and social benefits Channel 1 reverses when private and social benefits are negatively correlated, so that the relative fuel price signal is negative ($\pi' < 0$). If the clean fuel is privately more expensive and adoption occurs only because of non-usage benefits, WTP is decreasing in α_i while abatement remains increasing in α_i: status buyers who rarely drive adopt an EV first, while habitual commuters who would displace the most gasoline do not. A distortion then excludes the highest-α_i adopters at the margin—precisely those with the greatest abatement—so the fixed-subsidy margin has *lower* value than in the undistorted market (Figure B8). The elasticity channel can still favor fixed subsidies, so the net effect is empirical: Section 7 examines it in our context, and Appendix IX derives the reversal formally.

Open questions The model above could accommodate for example a budget neutral policy if environmental tax revenues must be earmarked for an accompanying environmental subsidy. Heterogeneity in γ across agents could shift the optimal instrument mix towards marginal pricing as this increases targeting on the externality (the counterfactual estimation in Section 7 assumes γ_i are distributed logit-normally). Solving explicitly for the optimal instrument mix s_1^*, s_2^*, τ^* as a function of the distribution of γ could offer a threshold beyond which distortions are sufficiently large that marginal pricing alone is no longer optimal. Loss aversion could play a role if agents weight losses (taxes) more than gains (subsidies): extensions could explore the efficacy of carbon taxes under distortions and estimate how large this asymmetry would need to be to generate the muted responses to marginal subsidies observed in many papers.

3 Study setting and experimental design

More than two billion people currently lack modern cooking technologies (IEA, 2024), and 73% of African households use charcoal or firewood as their primary cooking fuel (Berkouwer and Dean, 2026c). At the same time, more than two billion people have gained access to electricity in the past two decades (World Bank, 2024), generating widespread policy interest in 'leapfrogging' from biomass to electric cooking. East Africa is at the forefront: Kenya's electric utility launched the "Pika na Power" campaign in 2017, and the government's National Electric Cooking Strategy Action Plan followed with VAT waivers, support for consumer financing, and a dedicated e-cooking tariff (Section IV.1). Electric cooking is a particularly effective climate mitigation strategy where grids are clean: renewable sources generated 85% of Kenya's annual on-grid electricity in 2023 (Kenya Power, 2023).

64% of Kenyan households use firewood or charcoal as their primary cooking fuel (KNBS, 2024). The most common biomass stove in Kenya is a jiko, shown on the left in Figure 3. As a back-of-the-envelope calculation, at baseline respondents use on average 3.1 kilograms of charcoal per day, the production and burning of which emits approximately 4.9 tCO_2e per year (Table A1 presents household characteristics; Section 3.5 discusses the conversion from kilograms of charcoal to tons of CO_2e emitted).

It is worth emphasizing the scale of cooking-related emissions. The annual emissions of one cookstove in Kenya are roughly the same as the average annual emissions of a U.S. passenger vehicle, which is driven 11,500 miles per year at 22.2 miles per gallon of gasoline, emitting 8.9 kilograms of CO_2e per gallon or 4.6 tons of CO_2e per year (EPA, 2025).

Figure 3: Charcoal stove and induction stove

Notes: On the left is a charcoal stove instrumented with a high-frequency temperature sensor. On the right is the induction stove we study with the two pots and the frying pan that are included in the purchase bundle.

3.1 The ECOA induction stove

The right panel in Figure 3 shows the induction stove we study, manufactured by ECOA in Ruiru, Kenya. The induction stove has a power rating of 200W–2000W depending on the desired functionality (from low simmer to high boil) and a voltage rating of 240–250VAC and 50–60Hz. The median ECOA induction stove is in use for 2.2 years, though technology abandonment could be due to a combination of factors, including not only breakage but also non-payment, which shuts down the stove remotely (Figure B2 shows the survival curve).

At the time of our study launch in August 2025, more than 50,000 induction stoves had been sold, primarily through women's groups and pilot programs implemented through their sales agent programs in Tanzania and Kenya. That said, as the stoves were not yet available in retail outlets in our study area, our study population was generally not familiar with the product. The stove with one pot was priced at US$72 in Kenya. We offer participants a bundle consisting of the stove plus two pots and one frying pan, priced at US$82. In this context the price equals the production cost, so throughout the paper US$82 is both the stove price p and the cost of the technology.

Domestic residential customers in Kenya face an increasing block tariff. At the start of our study, customers consuming on average less than 30 kWh, 30–100 kWh, or more than 100 kWh per month, respectively, paid US$0.166, US$0.20, or US$0.22 per kWh.[9] In our sample, 86% consume less than 30 kWh per month and 99% consume less than 100 kWh per month. We use the lowest tariff band in calculations; the resulting estimates align well with estimates that use self-reported electricity spending data.

The company uses relatively sophisticated credit technologies to help alleviate credit market failures (Section IV.2 provides more detail). To cover these administrative costs, buyers paying in installments under the standard commercial plan must pay back 175% of the principal: an initial down payment of US$15 is followed by six monthly payments of US$19 each for a total price of

[9] Kenya's electricity tariff consists of a fixed component plus a floating component pegged to international fuel prices. The total price per kWh for customers in the top band fluctuated between US$0.2129 and US$0.2219 over the eight-month study duration.

US$132. Respondents could repay their loans early if they wished, but this would not reduce the total payment amount. The payment structure corresponds to an annualized percentage rate (APR) of 224%, well above Kenya's central bank rate of 9.75% during the experiment (CBK, 2026).

3.2 Data collection

We implement research activities in peri-urban neighborhoods around the cities of Naivasha, Gilgil, and Nakuru in Nakuru county in Kenya. Nakuru County is relatively representative of Kenya in terms of its fuel mix: 63% of households cook with firewood or charcoal and 65% have access to electricity, compared with 64% and 58% nationwide, respectively (KNBS, 2024). Respondents are visited up to four times (Figure A3 presents a timeline).

Baseline survey Study enumerators enrolled 2,503 people while walking around the study neighborhoods (Table A1 presents summary statistics). During enrollment, enumerators asked a small number of demographic and socioeconomic questions. To qualify for study participation, respondents had to own a standard Kenyan stove (shown on the left in Figure 3); spend at least half of their cooking fuel budget, and at least US$4 per week, on charcoal; be at least 18 years of age; have a pre-paid Kenya Power electricity meter in their homes; have a household size of at least 3 individuals; have basic literacy; and not currently own an electric or energy efficient charcoal stove.

Enumerators also installed temperature sensors on households' charcoal stoves. These would continue collecting 2-minute interval data continuously (Section IV.3 provides more detail).

Main survey 2,134 respondents completed the main survey, which was conducted around 50 days after the baseline visit (since attrition at this stage occurred before randomized assignment was revealed, it should not affect outcomes). Enumerators implemented the random treatments, provided (randomized) information about the induction stove, and elicited willingness-to-pay (WTP) using a Becker et al. (1964) mechanism (Section IV.4 provides more detail).

Respondents whose WTP was at least their randomly assigned price were expected to purchase the induction stove during the main visit (98% complied). They were then offered a standard 30-minute training on how to use the stove and how to contact customer service representatives in their area. Upon installation, the electric stoves record power usage in kilowatt-hours (kWh) on a 15-minute interval basis. This is transmitted wirelessly to remote servers in real-time, allowing us to subsidize the electricity used for the induction stove. The household surveys also ask about electricity spending, which we convert to usage using the US$0.166 per kWh electricity tariff.

Endline surveys 2,081 respondents completed the endline survey, conducted on average 40 days after the main survey.[10] Enumerators also briefly visited respondents around six months after the initial sensor deployment to download temperature data.

[10]Stove adopters are 2.2 percentage points less likely to attrit, but attrition is otherwise balanced by random treatment assignment (Table B2). 68% of attriters had moved outside the study area (Table B3).

Loan repayment For respondents who paid for the stove in installments, we observe daily payment data. In addition to measuring repayment, this also reveals the days on which the respondent is 'locked out' of their stove: this happens if they fall behind on their payments by at least one week (Section IV.2 provides more detail).

Market price survey We collect a panel dataset of charcoal prices by conducting market surveys over the course of the experiment across all three cities. This allows us to convert household charcoal expenditure data to kilograms of charcoal purchases.

3.3 Randomized treatments

Between the baseline and main surveys, each respondent is assigned to several randomized treatments (Figure A4 presents the randomization tree). All treatments are fully cross-randomized and stratified. Treatment assignment is furthermore stratified by baseline LPG ownership, baseline electricity expenditure, and baseline charcoal expenditure, and is balanced on observable baseline characteristics (Table B1).

Fixed-cost subsidy Participants were cross-randomized into different subsidy amounts for the fixed cost of buying the induction stove bundle, which included the stove and two pots and frying pan, shown in Figure 3. 49% of participants received a discount of 10% (for a price of US$73) and 46% received a discount of 75% (for a price of US$20) from the US$82 price.[11]

Marginal-cost subsidy Participants in the usage subsidy treatment groups receive a subsidy on the cost of the electricity that is used to operate the induction stove. One-third of participants do not receive a marginal-cost subsidy, one-third receive a subsidy of 25%, and one-third receive a subsidy of 75%. During the main visit, prior to the WTP elicitation, recipients are informed of the level and duration of subsidy they would receive and that only electricity used by the electric stove is subsidized. For a sense of magnitude, given realized usage averaging 15 kWh per month at a tariff of US$0.166, total marginal-cost subsidy payouts over the six-month subsidy period averaged US$3.7 and US$11 for respondents in the 25% and 75% subsidy groups, respectively.

Subsidies are paid out weekly over SMS using a unique, non-transferable token ID (Section IV.5 provides more detail). This mirrors the method and frequency of how households normally pay for electricity and is also similar to how the electric utility normally implements subsidy programs.

Payment structure One half of participants were randomly assigned to pay up front and one half to pay with an interest-free loan. Participants paying up front had to pay the full (randomly

[11] The remainder of participants had discounts of 46% (0.2% of participants), 50% (2.5%), or 69% (1.8%). All prices had positive probability of being selected from the feasible range of US$20 (since a down payment of US$12 plus at least some repayment was required by the stove firm) to US$73 (for ethical reasons all study participants received a discount of at least 10%).

assigned) price during the main visit. Participants paying with a loan had to pay a down payment of US$12.[12]

For those paying with a loan, starting one month after the purchase date the total amount due increases daily for 90 days. If a respondent did not meet the amount due, the stove would shut off seven days later until they made another payment, at which point the amount due would be recalculated from the new payment date (Section IV.2 provides more detail). Respondents could pay early but this would not reduce the total amount owed.

Information treatment All respondents receive two information sheets that show what meals they can cook with a charcoal stove and with an electric stove (Figure A5). 86% of participants are assigned to an information treatment group: their sheets list the cost of the electricity used to cook various meals on an induction stove and on a charcoal stove. For participants in the usage subsidy groups, the information sheets show costs with and without subsidy amounts.

3.4 Relevance to the model

The context described above closely matches the model presented in Section 2. To start, we observe an extremely inelastic intensive margin. As Section 4.4 will discuss, we observe no statistically significant drop in total time spent cooking (if anything, the point estimate is a 9% increase): the 27 minute per day decrease in charcoal cooking time is offset by a 37 minute increase in electric cooking time with very little simultaneous stove use (columns 7–9 of Table A2). A randomized SMS experiment implemented between 18 and 24 weeks after adoption furthermore shows very little movement in stove usage.

3.5 Converting energy usage to CO_2e emissions

Charcoal We assume a fraction of non-renewable biomass (fNRB)—the fraction of used biomass that is permanently removed, as opposed to forest that regrows naturally—of 38%, reflecting the most recently modeled fuelwood estimates for Kenya (Ghilardi and Bailis, 2024; UNFCCC, 2023). Still, our conclusions and results are relatively stable even when assuming an fNRB of 10% or 100% (Section 5.1). We convert estimates of production, processing, and combustion emissions of CO_2, CH_4 and N_2O to CO_2-equivalent emissions assuming a 50-year time horizon. Section IV.6 provides more detail.

Electricity Renewable sources of energy generate 85% of Kenya's annual grid electricity (Kenya Power, 2023), including geothermal (45%), hydroelectric (19%), and wind (17%). To convert the remaining 15% of generation to CO_2e emissions we assume an emissions factor of 1 tCO_2e per MWh of electricity, resulting in an average emissions intensity of 154 gCO_2e per kWh. While fossil fuel

[12] On the first two days of the main survey, September 23 and 24, a down payment of US$20 was used due to a miscommunication with the study partner. The exclusion of surveys collected on these days (which comprise 5% of the sample) does not meaningfully affect outcomes.

generation meets short-term demand, capacity expansions in recent years have all been renewable: average grid emissions are therefore a relevant margin in the clean cooking transition. We also report outcomes if Kenya's grid had the average emissions intensity of the U.S. grid.

Liquefied Petroleum Gas We convert LPG expenditures to cylinders purchased using local prices, and convert cylinders to CO_2e emissions using a conversion ratio of 4.6 kilograms of CO_2e per kilogram of LPG refill.[13]

Firewood 17% of households report ever using firewood to cook, but we do not track the quantity of wood collected nor firewood prices. Due to uncertainty in its origin, we refrain from converting the reduction in firewood usage to CO_2e emissions or estimating any CO_2e abatement generated by fuelwood reductions. Including this abatement would make the cost estimates more favorable.

4 Private and social impacts of electric stoves

Of the 2,134 respondents who successfully completed the main visit, 626 bought the induction stove, including 421 with a loan. Since stove purchasing itself was not randomly assigned, we use the randomly assigned subsidy, the randomly assigned loan access, and their interaction as instrumental variables to estimate the causal impacts of induction stove adoption on private and social outcomes (Section V presents these outcomes in the context of the pre-analysis plan). This consistently yields an F-statistic above 40.[14]

4.1 Charcoal stove usage

We convert temperature sensor data to an hourly panel dataset of cooking time by respondent, spanning from 45 days before to 90 days after the main visit. Figure 4 plots estimates of separate coefficients of the daily difference between adopters and non-adopters from 45 days before the main visit until 90 days after the main visit. Column 1 of panel a in Table 1 shows that daily time spent cooking using a charcoal stove, as measured by the temperature sensors, decreases by 31 minutes per day (Table A2 shows additional robustness checks). This corresponds to a 34% reduction in daily cooking time caused by adoption of the electric stove.

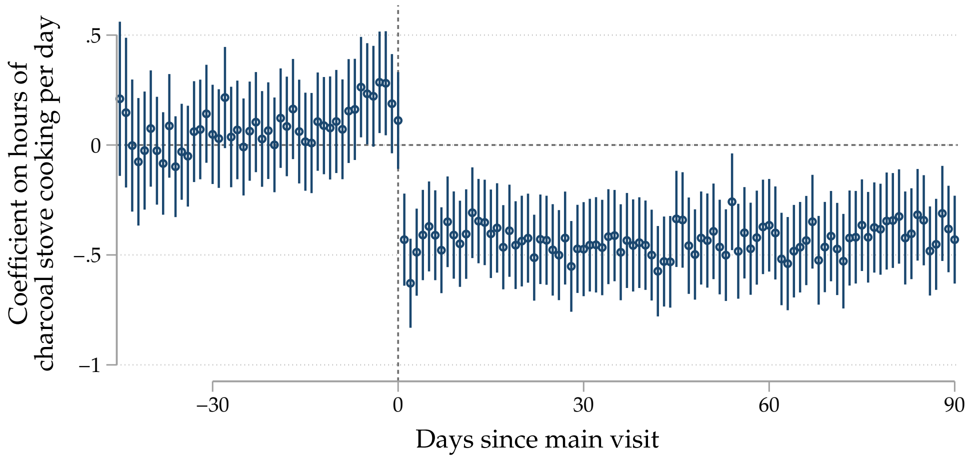
4.2 Energy expenditures

Table 1 presents estimates of the causal effect of stove adoption on energy expenditures. Self-reported monthly charcoal expenditures, shown in column 2 of panel a, decrease by US$10 (66%).

[13] One refill of a 6 kg canister of LPG in East Africa emits 27.8 kg CO_2e, or 4.6 kg CO_2e per kg LPG (USAID, 2021). This includes 3.0 kg CO_2e during combustion, 1.2 kg CO_2e during production, and 0.4 kg CO_2e during transport and raw materials production. A typical 6 kg LPG canister in Kenya costs approximately Ksh 1,400.

[14] 56% of respondents with a high subsidy and access to a loan bought the stove, whereas 1% of respondents with a low subsidy and paying up front bought the stove. 23% of respondents with a low subsidy and access to a loan bought the stove. 39% of respondents with a high subsidy but no access to a loan bought the stove.

Figure 4: Charcoal stove usage from Stove Use Monitors



Notes: Coefficients from a fixed effects regression using hourly data in which the outcome variable is the fraction of that hour the respondent was cooking (derived from the temperature data using the detection algorithm described in Section IV.3), multiplied by 24 for ease of interpretation. The regression includes date, event date, and hour of day fixed effects. Standard errors are clustered by respondent. Estimates from regressions with data collapsed to pre- and post-period differences in Table A2.

This is estimated from all compliers in the instrumental variables approach. Thus, it factors in behaviors like 'stacking' (where households continue to use their old stoves) as well as households choosing not to use their stoves. Without those behaviors, the estimated charcoal usage reduction would likely have been higher.

Column 5 shows that self-reported monthly electricity expenditures increase by US$2.3 (65%) per month.[15] This is very close to the value of the induction stoves' average consumption of 15 kWh per month, which equates to US$2.5 per month at the Kenya Power tariff of US$0.166 per kWh. There are additional, smaller decreases in wood and LPG expenditure. Aggregate monthly energy expenditures decrease by US$8.9 (36%). Assuming a 10% annual discount rate, this corresponds to US$209 over the stove's 2.2 year expected usage duration.

At the stove's price of US$82, the savings generate a return on investment of 118% per year. When paying with the commercial installment plan described in Section 3.1 the return is 81%. While these returns are substantial, they are less than the payment plan's 224% APR. For most households, the monthly energy savings do not cover the monthly repayment requirements, rendering adoption with credit prohibitively expensive for many. For comparison, the high and low subsidy pay plans in the experiment required monthly payments of US$3 and US$21, respectively.

The only baseline socioeconomic characteristic along which we see treatment effect heterogeneity is ownership of an LPG stove: non-owners save US$12 per month while owners save US$8 per month (Table B11). The p-value for this comparison is 0.095 but does not survive a multiple hypothesis testing adjustment.

[15]To address the confounding effect of the electricity subsidies, column 5 defines total electricity spending as self-reported electricity spending plus subsidy receipts. Column 6 estimates self-reported electricity spending among the subsidy control group.

Table 1: Impact of induction stove purchase (instrumental variables)

(a) Energy usage

	Hours per day cooking with charcoal	Monthly expenditures (USD)					
		Charcoal	LPG	Wood	Electricity		Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Bought induction stove (=1)	-0.52***	-10.01***	-0.85	-0.33	2.30***	2.18***	-8.89***
	(0.16)	(0.82)	(0.56)	(0.21)	(0.38)	(0.66)	(1.06)
Observations	5301720	2081	2081	2081	2081	703	2081
Control Mean	1.52	15.11	5.52	0.74	3.51	3.72	24.89
F-Stat		241.34	241.34	241.34	241.34	45.32	241.34

(b) Emissions (tons of CO_2e per year)

	Charcoal	LPG	Electricity	Total
	(1)	(2)	(3)	(4)
Bought induction stove (=1)	-2.53***	-0.03	0.02***	-2.53***
	(0.20)	(0.02)	(0.00)	(0.20)
Observations	2081	2081	2081	2081
Control Mean	3.77	0.17	0.04	3.99
F-Stat	122.03	122.03	122.03	122.03

Notes: Instrumental variables estimates use the randomized treatment price, the randomized credit condition, and their interaction as exogenous instruments. LPG is liquefied petroleum gas. Panel a shows the impact of induction stove purchasing on daily cooking and monthly energy expenditures. Column 1 uses hourly temperature sensor data (standard errors clustered by respondent and by date; additional specifications in Table A2). Column 5 adds subsidies to electricity expenditure. Column 6 estimates self-reported electricity spending among the subsidy control group. Panel b shows the impact on emissions. Section 3.5 describes the methodology for converting energy expenditures to emissions. 'Control mean' evaluated among non-adopters in the complier sample.

4.3 Emissions

We convert energy expenditures to CO_2e emissions using the methodology described in Section 3.5. While electricity expenditures constitute approximately 36% of total household energy expenditure among those who bought the stove, they constitute less than 3% of their baseline household emissions.[16] The increase in electricity usage thus generates grid emissions of 0.02 tCO_2e per year. For comparison, annual emissions from charcoal cooking average 4.9 tCO_2e (250 times more) and from LPG cooking average 0.3 tCO_2e (15 times more) (Table A1). Induction stoves as a technology are highly efficient in converting energy to heat. While Kenya's 85% renewable grid contributes to the low emissions, even at U.S. average grid emissions intensity the emissions from electric cooking would only amount to 0.06 tCO_2e per year.

Panel b of Table 1 shows an annual emissions reduction of 2.5 tCO_2e. This is effectively equivalent to the reduction in charcoal-related emissions, since the modest increase in electricity-related emissions is offset by a modest decrease in LPG-related emissions (as discussed in Section 3.5, we

[16] These numbers exclude transportation-related emissions or other emissions generated outside the home.

do not quantify fuelwood reductions).

4.4 Co-benefits and constraints

We briefly discuss the impacts of electric cooking on diet and health, and how power quality might affect adoption, below. These co-benefits and constraints are important from the perspective of poverty reduction and welfare improvement. However, they only affect the subsidy cost per tCO_2e to the extent that they move WTP and usage directly, which our measurements already incorporate. They matter for a social planner whose decisions incorporate welfare gains; however, even then they remain difficult to quantify. We therefore do not include them in the welfare calculations. Berkouwer and Dean (2026c) presents additional detail on the drivers and impacts of electric stove adoption that relate to health, socioeconomic status, and electrical grid quality.

Diet Induction stove adoption generates a 12 percentage point decrease in the fraction of people who cook ugali, mukimo, or matoke (three starch staples) and an 11 percentage point increase in the fraction of people who cook rice (Berkouwer and Dean, 2026c). Anecdotally, rice is easier to cook on an electric stove than other starches. We observe no other meaningful changes in which foods people cook nor the number of unique foods they cook. We do not observe any time savings: the 27 minute per day decrease in charcoal cookstove usage is more than offset by the 37 minutes per day of electric cooking. Given that respondents use both stoves simultaneously only 0.7% of cooking time, this leaves households cooking 8.2 minutes more per day (columns 7–9 of Table A2).

Health We see meaningful reductions in self-reported cough and breathlessness among respondents and their children. In a similar context, Berkouwer and Dean (2026a) found health improvements of similar magnitude. While these are potentially important for welfare, those improvements did not generate benefits in terms of long-term clinical health or economic outcomes like hours worked in that study.

Power quality In a similar context in Kenya, Wolfram et al. (2027) record an average of 61 minutes of power outage per day, as well as imperfect voltage quality: average voltage of 233V is 3% lower than nominal voltage of 240V. This could have an important dampening effect on the value of electric cooking in this context.

5 Green subsidy efficacy

5.1 Fixed-cost subsidies

We first evaluate the subsidy cost of incentivizing a marginal adoption by leveraging the demand curve elicited through the BDM exercise (panel a of Figure 7). The high price is the competitive market price minus a small subsidy $p - s_l$. A larger subsidy lowers the price further to $p - s_h$, such that $0 \leq s_l < s_h$. Denote q_l and q_h to be the fraction of individuals who would buy the stove with the low or high subsidy, respectively. The subsidy cost required to incentivize one additional adoption can then be denoted as follows:

$$\mathbb{C}_s = \frac{\text{Additional subsidy cost}}{\text{Additional adoptions}} = \frac{q_h s_h - q_l s_l}{q_h - q_l} \tag{3}$$

In the context of this study, absent access to a loan 40% of respondents would buy the stove with the high subsidy whereas 2.6% of respondents would buy the stove with the low subsidy. Applying Equation 3, the principal spends $\mathbb{C}_s =$ US\$65 in subsidy expenditures to incentivize one additional adoption.

Subsidy efficiency can then be calculated by combining this with the stove price $p =$ US\$82 and the benefits per additional adoption. Marginal buyers of induction stoves in our context save US\$9.7 per month and abate 2.6 tCO_2e per year. Discounting at 10% per year, this gives a discounted lifetime private benefit per marginal adopter of $\underline{\hat{b}} =$ US\$228 and lifetime abatement of $\underline{\hat{\phi}} = 5.7$ tCO_2e (the total discounted monthly savings and annual abatement coefficients in columns 8 and 10 of Table 4 after 2.2 years). Using a Social Cost of Carbon (SCC) of US\$120 (the 2020 Environmental Protection Agency estimate using a 2.5% discount rate; EPA, 2023), this corresponds to a marginal externality benefit of $\text{SCC} \cdot \underline{\hat{\phi}} =$ US\$688.

It is worth noting that the causal estimates from the instrumental variables (IV) regression were identified off of the compliers, who have a willingness-to-pay (WTP) between the low and high prices ($p - s_h =$ US\$20 and $p - s_l =$ US\$73). The local average treatment effect is thus estimated for adopters who are marginal to an increase in the subsidy from s_l to s_h. Because compliers in the IV estimation are exactly the adopters marginal to the subsidy increase, we can conduct several relevant policy calculations.

The results below present estimates for the subsidy cost per tCO_2e and the observable welfare gain per subsidy dollar, reflecting the objective functions of the environmental principal and the social principal, respectively. Table B7 defines and presents results for two additional measures: marginal value of public funds and resource cost per ton of CO_2e abated.

Subsidy cost per tCO_2e We first consider an environmental principal who wishes to maximize CO_2e abatement while minimizing subsidy expenditure.[17] This could be a budget-constrained government agency or a cost-minimizing company or organization that wishes to meet an abatement target. The additional subsidy expenditure required to abate one additional ton of CO_2e is then given by:

$$\text{Subsidy cost per } tCO_2e = \frac{\text{Subsidy cost per additional adoption}}{\text{Tons of } CO_2e \text{ abated per adoption}} \tag{4}$$

Applying the estimates from above, subsidies for induction stoves in this context have a subsidy cost per tCO_2e of US\$11.

[17] The decision of whether to express efficiency as abatement per dollar or dollars per abatement is arbitrary; we use subsidy cost per tCO_2e to facilitate natural intuition with the abatement cost curve and offset market prices.

Observable welfare gain per subsidy dollar Since WTP is distorted in this context, we cannot use it to measure consumer surplus. We instead evaluate the observable components of welfare per subsidy dollar, setting unobserved switching costs to zero. In the case of a pure allocation distortion, this consists of fuel savings plus the value of the externality avoided minus the production cost of the improved stove (conservatively, we assume the cost of the traditional stove is 0):

$$\text{Benefit per subsidy dollar} = \frac{\text{Benefit per marginal adoption}}{\text{Subsidy cost per marginal adoption}} = \frac{\underline{\hat{b}} + \text{SCC} \cdot \underline{\hat{\phi}} - p}{\mathbb{C}_s} \quad (5)$$

This yields an observable welfare gain of US\$12.8, consisting of US\$3.5 in private fuel savings and US\$11 in environmental benefit (minus US\$1.3 in stove cost), per subsidy dollar.[18]

A complete estimate of welfare impacts should also include the resource distortion costs, if any. Recall that a subsidy s_1 incurs $r(p - s_1)$ in additional resource cost for all new adopters, but also lowers resource cost for inframarginal adopters by rs_1. Our preferred estimate sets $r = 0.5$ for respondents paying the full price up front (Section 6.3). This yields an observable welfare gain of US\$12.6 per subsidy dollar, close to the US\$12.8 obtained under a pure allocation distortion, indicating that the resource cost is small relative to the fuel savings and abatement generated by adoption. Most conservatively, we could treat the entire wedge as a resource distortion. This corresponds to an implied resource rate of $r = 973\%$ and a welfare gain of US\$10.3 per subsidy dollar. A resource distortion of this magnitude is unlikely, but the resulting estimate is a useful lower bound on the welfare benefit.

This subsidy cost per tCO_2e is substantially lower than that of many other green technologies. Panel a of Table 2 summarizes the subsidy cost per tCO_2e estimated in this paper as well as in other papers evaluating energy efficient technologies in contexts where adopters likely face significant demand distortions. Panel b lists estimates of green technologies receiving subsidies in the U.S. By and large, subsidies for green technologies in low- and middle-income contexts generate larger social benefits per subsidy dollar spent than similar subsidies for green household technologies in high-income contexts.

These differences are partly driven by the underlying technologies. For example, each US\$82 stove abates on average 2.5 tCO_2e per year while a US\$40,000 electric vehicle abates 0.1–1.5 tCO_2e per year relative to the counterfactual (Muehlegger and Rapson, 2023). However, even conditioning on technology, the channels identified in this paper—differences in the marginal adopter and demand elasticity—drive increases in subsidy efficiency. Section 6 first disentangles how the two channels contribute to subsidy efficiency. Section 7 then estimates the model presented in Section 2 to disentangle how much demand distortions per se reduce the subsidy cost per tCO_2e relative to estimates from contexts with fewer demand distortions.

[18] This assumes an SCC of US\$120. Using an SCC of US\$190 (the 2020 number with a 2% discount rate) or US\$380 (the 2030 number with a 1.5% discount rate) increases the observable welfare gain to US\$19 or US\$36, respectively.

Table 2: Efficiency of selected green subsidies

Technology	Observable welfare gain per US$1 subsidy (US$)	Subsidy cost per tCO_2e (US$)	Source
(a) Low- and middle-income contexts			
Electric stoves	US$12.6	US$11	*This paper*
Solar lanterns	12	10	Rom et al. (2023)
Energy efficient motors	4.1	-[a]	Chaurey et al. (2025)
Improved charcoal stoves	322[b]	6	Berkouwer and Dean (2026b), Hahn et al. (2026)
(b) High-income contexts			
Electric vehicles	0.4	1,356	Hahn et al. (2026)
Electric vehicles	0.3	1,202	Allcott et al. (2026)
Wind production credits	4.9	46	Hahn et al. (2026)
Residential solar	2.9	90	Hahn et al. (2026)
Appliance rebates	0.2	474	Hahn et al. (2026)
Vehicle retirement	0	876	Hahn et al. (2026)
Hybrid vehicles	0	5,940	Hahn et al. (2026)
Weatherization	0	779	Hahn et al. (2026)

Notes: We conservatively assume a social cost of carbon (SCC) of US$120; Hahn et al. (2026) assume US$193. [a]Chaurey et al. (2025) estimate the impact of disseminating energy efficient motors for free to non-adopters. We were unable to calculate the subsidy cost per tCO_2e from the numbers provided in the manuscript. [b]The US$322 benefit estimated by Berkouwer and Dean (2026b) results from the low technology cost (US$40), large emissions reductions (2.1 tCO_2e per year), large financial savings (US$78–114 per year), and a long stove lifetime (2.7 years).

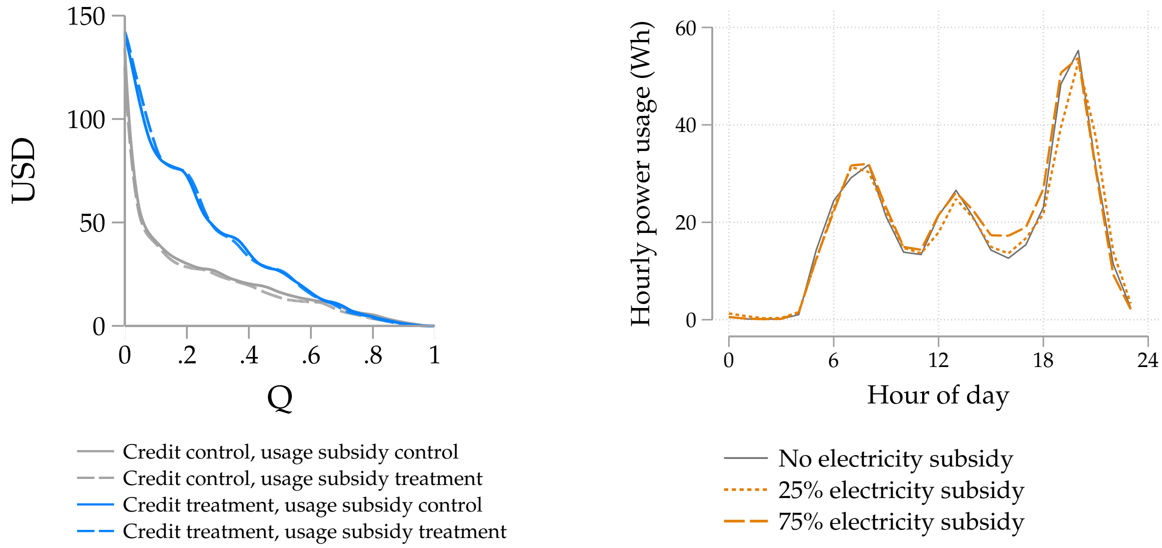
Results under alternative assumptions The benchmark scenario evaluated in this paper uses 2.2 years of stove usage (based on the company's historical data, shown in Figure B2) and uses Kenya's average grid emissions. Alternatively, assume that the stove's lifespan is on average 1 year or 3 years. This changes the subsidy cost per tCO_2e among respondents who were not also given access to a loan to US$25 and US$8, respectively (scenarios 1 and 2 of Table A3).

There is significant uncertainty around the fraction of non-renewable biomass (fNRB). Our primary estimates use the 38% rate that Ghilardi and Bailis (2024) estimate using their Modeling Fuelwood Savings Scenarios (MoFuSS) model, per the recommendations of Article 6.4 of the Paris Agreement. To measure the sensitivity of our results to this parameter, we also calculate costs under 10% and 100% rates. This changes the subsidy cost per tCO_2e to US$14 or US$8.4, respectively (scenarios 3 and 4 of Table A3). These differences are relatively minor because most charcoal-related emissions are incurred not during combustion but during production and processing, which are not affected by the assumed fNRB (Floess et al., 2023).

Finally, Kenya's electricity generation mix is 85% renewable, generating less than half the emis-

Figure 5: Impacts of marginal-cost subsidies

(a) On willingness-to-pay (extensive margin) — (b) On induction stove usage (intensive margin)



Notes: Panel a shows the impact of access to credit and electricity subsidies on demand, with demand curves constructed using WTP values elicited through the Becker-DeGroot-Marschak mechanism (point estimates in Table A4). In panel b, electric stove usage in Watt-hours recorded by the induction stoves, separated by whether the respondent was in the electricity subsidy control group, the 25% electricity subsidy group, or the 75% electricity subsidy group (point estimates in Table A6). Figure B3 presents demand curves that adjust for observed repayment rates.

sions per kWh of the U.S. grid. However, even if Kenya's grid emissions were equivalent to the U.S. grid, the subsidy cost per tCO_2e would remain essentially unchanged (scenario 5 of Table A3). Cooking with charcoal is significantly dirtier than cooking with electricity, such that the large reduction in charcoal cooking emissions is only very slightly offset by the increase in electricity cooking emissions (as discussed in Section 4.3).

5.2 Marginal-cost subsidies

The absolute efficacy of fixed-cost subsidies does not inform their relative efficacy compared with marginal-cost subsidies, which could offer two advantages over a fixed-cost subsidy of equivalent net present value. First, marginal-cost subsidies could encourage selection into adoption by higher users. Second, they could incentivize switching conditional on adoption. Both channels could increase average usage levels conditional on adoption, increasing the efficiency of marginal-cost subsidies in terms of CO_2e abated per dollar of subsidy expenditure relative to an equivalent fixed-cost subsidy.

On the other hand, if respondents are myopic or imperfectly attentive to electricity expenditures, they might undervalue electricity subsidies that accrue in the future. This could undermine the efficacy of marginal-cost subsidies. The relative efficacy of these two subsidy instruments is thus an empirical question.

We evaluate the impact of the randomly assigned electricity subsidies, which lowered the cost of cooking by 25% or 75%, each assigned to one-third of the sample. Since we announced the electricity subsidy prior to eliciting WTP, the announcement of the usage subsidy could have affected usage both through a direct effect as well as a selection channel. Figure 5 therefore shows the impact of usage subsidies on both the extensive margin (purchasing the stove) and the intensive margin (using the stove). Panel a shows WTP by whether the respondent was given access to a loan and whether they received a marginal-cost subsidy. The electricity subsidy has no statistically detectable effect on average WTP. We can rule out that each US$1 in electricity subsidy has more than a US$0.2 impact (column 3 of Table A6). Pooled regressions and separate regressions by subsidy level all show no statistically significant impact on WTP (Table A4). The interaction term between credit and subsidies is positive, but this effect is not significant and very noisy: electricity subsidies may have a positive impact on WTP when respondents' credit constraints are relaxed, or credit causes selection by people who are slightly more price elastic. We are unable to cleanly evaluate whether this is the case in this context.

This result is what the model predicts at the distortions estimated in this context. On the extensive margin, a marginal-cost subsidy raises perceived WTP at only the pass-through rate $1 - \gamma$. Taking the ratio of WTP to discounted fuel savings, we estimate a demand distortion of 0.91 (Section 6). Each US$1 of electricity subsidy should raise WTP by only US$0.09. This predicted pass-through lies inside our confidence interval, while undistorted one-for-one pass-through is rejected. Viewed through the model, the unresponsiveness of WTP to electricity subsidies is an implication of the same distortion between WTP and fuel savings.

Panel b of Figure 5 presents average hourly induction stove usage by electricity subsidy treatment group. We estimate a demand elasticity of -0.03, and we can rule out that a 1% increase in price causes more than a 0.12% decrease in usage (column 6 of Table A6). The point estimate for the 75% subsidy is larger than the point estimate for the 25% subsidy but less than 10% of mean usage and not statistically different from zero.

We explore two explanations for the unresponsiveness to electricity subsidies on the intensive margin in this context. First, households may perceive a hassle cost in topping up a prepaid meter: recall that recipients had to manually enter a unique code on their electricity meter to activate the subsidy each week. However, this is Kenya Power's standard method for buying electricity, which households in our survey sample do on average once per week anyway—the same frequency at which we transfer electricity subsidies.

Alternatively, cooking habits may simply be very inelastic to price. To evaluate this, we implement a randomized SMS experiment over a period of two weeks between 18 and 24 weeks after respondents purchased the electric stoves. Each respondent received one SMS at 2pm on four randomly selected days (half of respondents received them all during the first week, the other half during the second week). The SMS reminded them of the cost of cooking a specific meal, which was tailored according to their marginal cost. For those in the marginal-cost subsidy treatment groups, it reminded them that they were getting a subsidy. Across a host of specifications, the SMS messages had no detectable impact on electric cooking patterns (Figure A7 shows identical daily consumption patterns; Table B5 presents empirical estimates using hourly data). This evidence is in line with an inelastic intensive margin due to preferences: respondents might have strong preferences over which meals to cook on electric stoves and which meals to continue cooking on charcoal stoves.

Together, these results indicate that fixed-cost subsidy instruments are more effective in this context than marginal-cost subsidies.

6 Demand distortions and subsidy efficacy

One of this paper's central findings is that the relatively low subsidy cost per tCO_2e discussed above is caused at least in part by demand distortions. Section 4 showed that the stove's average discounted fuel savings are $b =$US$209 after 2.2 years. Respondents paying up front have an average willingness-to-pay (WTP) of US$20. Defining WTP as $wtp = b \cdot (1 - \gamma)$, this suggests a demand distortion of $\gamma = 0.91$. Throughout, we compute γ using the pooled savings estimate (Table 1): the wedge is a property of decision-making, and defining it against a common benchmark keeps the two credit arms comparable. The observed WTP relative to the stream of savings corresponds to an implicit discount factor of 0.544 per month, or 0.0007 per year. Put differently, respondent decision-making reflects indifference between receiving US$1 today or receiving US$1,487 one year from today.

Figure 5 shows that access to credit in this context increases WTP by 81% (column 4 of Table A4). Respondents given access to a loan have an average WTP of US$35 suggesting a demand distortion of $\gamma_1 = 0.83$. By the end of data tracking, 271 days after stove sale on average, respondents who had bought the stove with a loan had paid on average 86% of the price (Figure A6). Accounting for observed repayment rates either increases or decreases the treatment effect in percentage terms depending on which outcome is used, but the effect is always large and statistically significant (Table B4). The discount factor increases to 0.0308, at which point US$1 today is worth US$32 in one year. The difference between these values illustrates the degree to which capital constraints distort decision-making in this context.

How do these demand distortions affect subsidy efficacy? Table 3 presents subsidy efficiency estimates for respondents whose demand was more distorted and less distorted. Its rows are the empirical counterparts of the model's summary statistics. The Channel 1 rows are the empirical counterparts of the average usage of induced adopters ($\bar{u}$), valued at the per-unit fuel saving and at the per-unit externality. The Channel 2 rows are the empirical counterparts of demand elasticity and the number of subsidy recipients per induced adoption (ζ). Proposition 1 signs both movements: selection raises $\bar{u}$ while higher elasticity lowers ζ, so usage per subsidy dollar rises on both counts. The two aggregate rows correspond to the objectives of the two principals. The social principal's observable welfare gain per subsidy dollar depends on the split between allocation and resource distortion, as characterized in Section 2.5.

We define the observable welfare gain as consisting of fuel savings and externality benefit minus stove production cost and the resource cost (if any). Under our preferred estimate, subsidizing green

Table 3: Impact of demand distortions on subsidy efficacy through the two channels

	Less distorted ($\gamma_1 = 0.83$)	More distorted ($\gamma_0 = 0.91$)	Advantage of subsidies in a distorted market
Channel 1: Marginal adopter			
Fuel savings per marginal adoption ($\hat{\underline{b}}$)	US\$175	US\$228	+30%
Abatement per marginal adoption ($\hat{\underline{\phi}}$)	4.8 tCO_2e	5.7 tCO_2e	+19%
Channel 2: Subsidy cost			
Extensive margin elasticity to fixed subsidy	$\mathcal{E} = -0.9$	$\mathcal{E} = -2.3$	
Subsidy expenditure inframarginal	38%	6.5%	
Subsidy expenditure per marginal adoption	US\$93	US\$65	−30%
Aggregate difference			
Subsidy cost per tCO_2e	US\$19	US\$11	−41%
Observable welfare gain per subsidy dollar			
Pure allocation distortion ($r = 0$)	US\$7.2	US\$12.8	+77%
Preferred estimate ($r = 0.5$)	US\$7.2	US\$12.6	+76%
Pure resource distortion ($r = \frac{\gamma}{1-\gamma}$)	US\$7.2	US\$10.3	+43%

Notes: How demand distortions affect subsidy efficacy. 'Less distorted' are respondents with access to an interest-free loan; they carry $r = 0$ throughout. 'More distorted' are respondents who pay the full price up front. Fuel savings and abatement per marginal adoption are totals after 2.2 years. Observable welfare gain consists of fuel savings plus externality benefit minus the economic cost of stove manufacturing and the resource cost (if any). Under an allocation distortion the wedge consumes nothing. Under a resource distortion each induced adopter incurs additional resource cost but the subsidy lowers the resource cost for inframarginal borrowers. The preferred estimate sets $r = 0.50$ (Section 6.3 derives this rate). The final row assumes the entire wedge is resource cost; at γ_0 this implies $r = 9.7$. Section 2.1 describes the theoretical underpinnings of channels 1 and 2. Section 6.1 and Section 6.2 present empirical estimates for channels 1 and 2. The GMM estimation in Section 7 uses the $\hat{\underline{b}}$ and $\hat{\underline{\phi}}$ estimates.

technologies for recipients with larger credit distortions generates 76% more observable welfare gain than doing so among recipients whose distortions have been relaxed, from US\$7.2 up to US\$12.6 per subsidy dollar. The result also holds for a principal who only cares about maximizing abatement per dollar of subsidy expenditure. The subsidy cost per tCO_2e of US\$19 is 70% higher than the US\$11 among those with a higher demand distortion.[19]

Section 6.1 and Section 6.2 quantify how the externality generated by the marginal adopter (channel 1) and demand elasticity (channel 2), respectively, drive these results.

6.1 Channel 1: Fuel reduction of marginal adopters

We first test the impact of demand distortions on the average externality of the marginal adopter. Throughout, we define a marginal adopter as having a WTP between the low price and the high price, so that they would adopt the stove if and only if there was an increase in the subsidy:

[19] Relaxing the distortion lowers the marginal value of public funds (MVPF) by 41% (from 14 to 8.2) and the resource benefit by 24% (from US\$25 to US\$19). Table B7 shows more detail.

Table 4: Monthly savings and annual abatement among marginal electric stove adopters

	Charcoal				Electricity		Total			
	(1) USD	(2) USD	(3) CO_2e	(4) CO_2e	(5) USD	(6) USD	(7) USD	(8) USD	(9) CO_2e	(10) CO_2e
Credit (=1)	-1.0	-1.0	-0.2	-0.2	0.1	0.1	-0.7	-0.7	-0.2	-0.2
	(1.4)	(1.4)	(0.4)	(0.4)	(0.8)	(0.8)	(2.0)	(2.0)	(0.4)	(0.4)
Bought stove (=1)	-10.5***		-2.6***		2.3***		-9.7***		-2.6***	
	(0.6)		(0.2)		(0.3)		(0.9)		(0.2)	
Bought X No credit		-10.5***		-2.6***		2.3***		-9.7***		-2.6***
		(0.6)		(0.2)		(0.3)		(0.9)		(0.2)
Bought X Credit	1.6*	-8.9***	0.4*	-2.2***	-0.2	2.2***	2.2*	-7.4***	0.4*	-2.2***
	(0.9)	(0.6)	(0.2)	(0.2)	(0.5)	(0.3)	(1.2)	(0.8)	(0.2)	(0.2)
Observations	754	754	754	754	754	754	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9	3.6	3.6	25.5	25.5	4.1	4.1

Notes: Instrumental variables regressions using the randomly assigned price as an instrument for adoption to estimate its impact on spending and emissions among marginal adopters (i.e., WTP $\in [20, 73)$ US$). USD amounts are monthly; CO_2e amounts are annual. IV compliers correspond to marginal adopters. Odd columns test whether the marginal adopters' average externality differs by credit condition. Even columns estimate the positive externality by credit condition. Endline surveys were conducted with 754 of the 765 marginal adopters (98.6%).

$p - s_h \leq wtp < p - s_l$. There are 387 marginal adopters in the credit control group and 378 marginal adopters in the credit treatment group for a total sample of 765 marginal adopters.

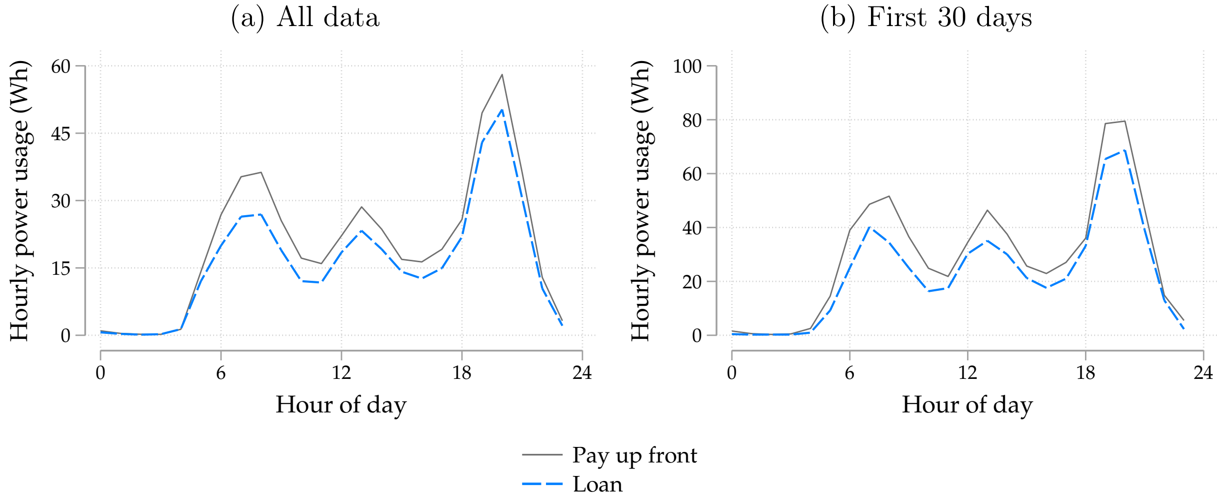
In the context of the model, when agents face larger demand distortions, agents who are marginal to a subsidy should unambiguously have higher average usage, and hence higher fuel savings (Lemma 2). Columns 7 and 8 of Table 4 offer a test of this. Marginal adopters with a larger demand distortion generate on average US$228 in discounted private savings over 2.2 years, whereas adopters whose demand distortion was reduced through access to credit generate US$175, a statistically significant difference.

Since social and private benefits are positively correlated in this context, this translates into a similar difference in positive externalities. Columns 9 and 10 of Table 4 show that this is the case. Marginal adopters in the credit control group abate 5.7 tCO_2e after 2.2 years whereas marginal adopters in the credit treatment group abate 4.8 tCO_2e. In other words, the credit constraints in this context increase the abatement caused by marginal adopters by 19%.

Figure 6 plots induction stove usage over the course of the day among adopters who were marginal to the subsidy, by credit treatment group. Marginal adopters in the credit control group have an average usage of 538 Wh per day, which is 18% higher than marginal adopters in the credit treatment group, who have an average daily usage of 455 Wh per day (Table A5 shows the estimates corresponding to panel a).

This difference could reflect a treatment effect rather than selection. For example, having to pay back a loan leaves less money available for electricity expenditures. However, panel b shows that these effects begin within the first month, when adopters paying up front would have just faced a large direct liquidity shock from adoption. If anything, this would bias the estimates toward

Figure 6: Electric stove usage among marginal adopters by credit condition



Notes: Hourly usage in Watt-hours by randomized credit condition among marginal adopters, defined as having $p - s_h \leq wtp_i < p - s_l$. Table A5 presents point estimates, which are significant with $p < 0.05$. The IV F-stat is 3,151 for all regressions.

finding less usage among adopters paying up front. The difference in usage across the two groups is therefore likely to be a result purely of selection rather than a treatment effect. Either way, the interpretation of this wedge does not affect the subsidy efficacy calculations.

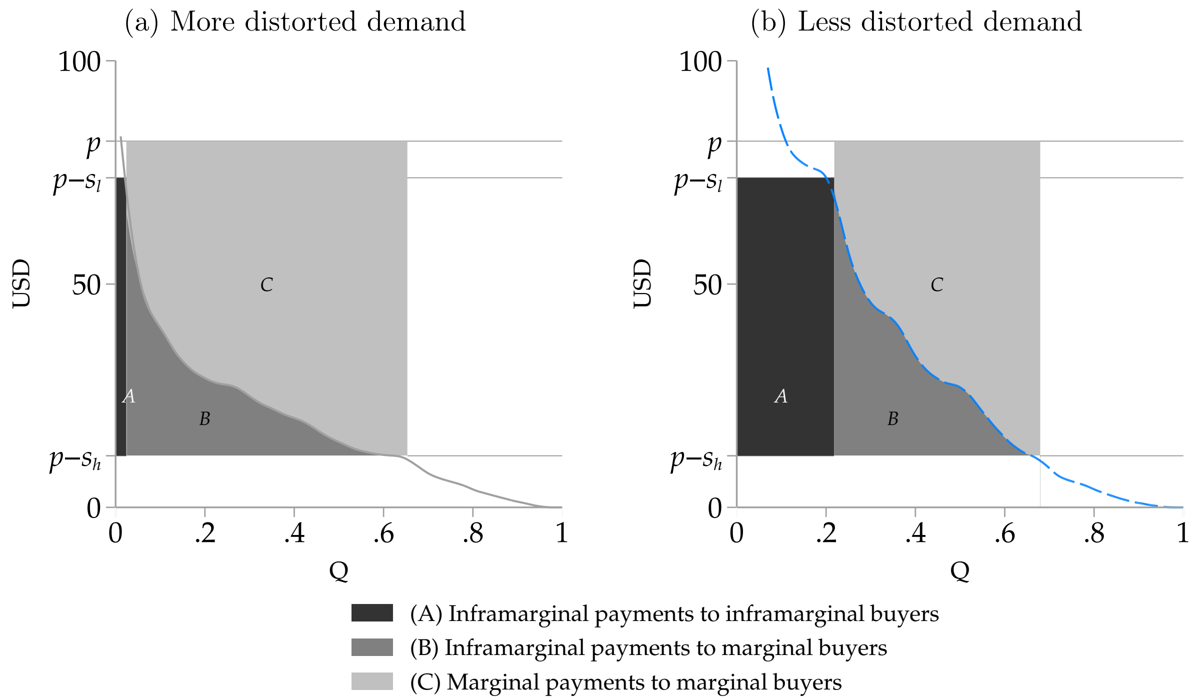
6.2 Channel 2: Subsidy expenditure per marginal adoption

The second channel through which demand distortions increase subsidy efficacy is by increasing demand elasticity, which both increases the number of agents who are marginal to the subsidy, and reduces the number of inframarginal payments. Distorted elasticity at any price equals undistorted elasticity at the scaled-up price, so the inframarginal cost per induced adoption falls with the distortion (Lemma 3). The cost increase resulting from a subsidy increase consists of subsidy expenditures towards both inframarginal and marginal buyers:

$$q_h s_h - q_l s_l = \underbrace{q_l(s_h - s_l)}_{\substack{\text{Increased payouts to} \\ \text{inframarginal buyers}}} + \underbrace{s_h(q_h - q_l)}_{\substack{\text{Full payouts to} \\ \text{marginal buyers}}} \tag{6}$$

Figure 7 presents intuition on how these effects operate in the context of the demand curves elicited through the BDM. Area A represents the first term in Equation 6 and $B \cup C$ represents the second term. Area A as a fraction of areas $A \cup B \cup C$ denotes the fraction of additional subsidy expenditures disbursed to inframarginal buyers. Comparing panels a and b, demand distortions in this context both decrease payouts to inframarginal buyers in absolute terms and as a fraction of total payouts. 2.6% of people in the credit control group would buy the stove at the high price whereas 21% of people in the credit treatment group would. As a result, among the credit treatment group 38% of subsidy expenditures flow to inframarginal buyers whereas in the credit control group

Figure 7: Marginal and inframarginal expenditures by credit condition



Notes: Demand curves constructed using WTP values elicited through the BDM mechanism. Respondents having to pay up front have more distorted demand than respondents randomly given access to a loan. Demand distortions increase the proportion of marginal buyers and decrease the proportion of inframarginal buyers that result from an increase in the subsidy from s_l to s_h.

only 6.5% of expenditures do.

Taken together, the environmental principal incurs a total subsidy cost of US$93 per additional sale among respondents with access to a loan, whereas respondents in the credit control group require additional subsidy spending of US$65 per additional sale. In other words, relaxing demand distortions increases the subsidy cost of achieving one additional sale by 43%.

These impacts can be summarized by the change in the demand elasticity, which as a percentage term reflects both marginal adopters (in the numerator) and inframarginal adopters (in the denominator). Table 5 estimates the price-quantity coefficients. Column 5 shows a demand elasticity of -2.3 among participants required to pay the entire price up front and column 6 shows -0.9 among people paying with a loan.

An alternative, policy-relevant measure of demand elasticity would be to take s_l as the policy baseline and then to estimate the percentage quantity and price change from this starting point. This results in a demand elasticity of -20 among the control group and -2.3 among respondents whose credit constraints were partially relaxed.

Columns 2 and 3 indicate that the increased elasticity includes an increase in the numerator: there are more marginal adopters per US$1 subsidy. Given that there are also fewer inframarginal adopters (the denominator in the elasticity calculation), column 5 shows that respondents with a larger demand distortion have an even larger demand elasticity.

Table 5: Extensive margin demand elasticities by credit treatment group

	Quantity (%)			Log(Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)
Price (USD)	-0.84***	-0.96***	-0.80***			
	(0.01)	(0.02)	(0.01)			
Log(Price)				-1.23***	-2.33***	-0.88***
				(0.01)	(0.02)	(0.01)
Observations	765	387	378	765	387	378
Sample		C=0	C=1		C=0	C=1

Notes: We estimate elasticities over the policy-relevant sample of compliers, i.e., anyone with willingness-to-pay between $p - s_h$ and $p - s_l$. Column 1 shows that a US$1 increase in price causes a 0.8 percentage point decrease in the quantity demanded. Columns 2 and 3 show that this decrease is steeper for respondents in the control group than in the treatment group. The lower rate of inframarginal adoption exacerbates the difference in elasticities in columns 5 and 6, as the small denominator among the credit control group causes a larger demand elasticity.

6.3 Evaluating resource cost

The welfare results in Table 3 depend on how much of the demand distortion consumes real resources. Most conservatively we can assume that the entire wedge represents resource cost. Applying $\gamma = \frac{r}{1+r}$ literally to the entire wedge implies $r = 9.7$; roughly US$200 of real resources per US$20 financed. While at least some of the wedge is likely an allocation wedge, we report the resulting US$10 to bound the exercise.

Respondents with a loan still value the stove at only 17% of its lifetime fuel savings. We interpret this residual as an allocation distortion—present bias, inattention to fuel savings, or misperceived benefits—common to both arms. We therefore attribute to financing only the incremental wedge that the credit randomization identifies, and charge $r = 0$ to respondents holding a loan. Writing the pass-through among respondents paying up front as the product of a common allocation component and a credit-specific one ($1 - \gamma = \frac{1-\rho}{1+r}$, as defined in Section 2.1), with ρ common to both arms and $r = 0$ for respondents holding a loan, the credit-attributable resource rate is $1 + r = \frac{1-\gamma_1}{1-\gamma_0}$, or $r = 81\%$.

Reassuringly, this is close to the rate the market charges: the commercial installment product for this device requires repaying 175% of the amount borrowed. Respondents in our sample repaid 86% of the amount they owed on average (Figure A6). Netting out the down payment and the implied default share (which are transfers) from the 75% interest, and assuming the stove company does not charge a markup, leaves administrative costs—hiring field agents to visit late payers, maintaining an online payment tracking system—and cost of funds together as roughly half of principal. We therefore adopt $r = 0.5$ as our preferred estimate for respondents paying up front. The resource cost equals $r \cdot (p - \mathbb{C}_s)$ per induced adoption: its sign therefore depends on whether subsidy spending per marginal adoption exceeds the stove price.[20] Among respondents paying up front it does not

[20]This is the resource cost incurred by the induced adopter net of the resource reduction that the larger subsidy delivers to inframarginal borrowers.

(US$65 against a price of US$82), so the term is a net cost of US$8.3 per induced adoption under the preferred estimate. Our preferred estimate is therefore an observable welfare gain of US$12.6 per subsidy dollar.

Finally, we assess the assumption of $r = 0$ in the less distorted column. The study offered interest-free loans. Among respondents with loan access, subsidy spending per marginal adoption is US$93 (38% of subsidy expenditure is inframarginal). Since this exceeds the price of the stove, the resource savings for inframarginal buyers outweigh the financing cost incurred by the induced adopter. Charging the loan arm the same rate as respondents paying up front would therefore raise its welfare gain and narrow the measured advantage of subsidizing the more distorted group, from 76% to 74% under the preferred estimate and from 43% to 32% under the literal reading ($r = \gamma_1/(1-\gamma_1) = 4.9$).

The result in Corollary 1—that welfare per fixed-cost subsidy dollar rises with both allocation and resource distortions—requires two conditions. The first is that induced adoptions are valuable enough: the fuel savings and abatement of the average induced adoption, net of the stove price, cover r times that price for any r below 10.2, comfortably satisfied at our preferred estimate of $r = 0.5$ (even the most conservative estimate of r sits inside the condition). The second is that the program spends at least the technology price in fixed-cost subsidies per adoption it induces. This holds for respondents with loan access, where US$93 exceeds the US$82 price. However, it fails for respondents paying up front. Signing the effect of the resource distortion for $r > 0$ thus instead requires an additional condition: that the rate at which welfare per subsidy dollar rises with the distortion exceed the direct resource cost of the induced adoptions, net of the financing the subsidy retires on inframarginal buyers. Differencing the two arms at a common $r = 0.5$, the former is 6.0 and the latter 0.38, so the condition holds by more than an order of magnitude.

7 Model estimation and counterfactuals

Even among respondents who are given access to a loan, willingness-to-pay (WTP) of US$35 is still well below the total savings of US$209 after 2.2 years (savings are discounted at 10% per year). It is reasonable to think that our short intervention did not fully alleviate all demand distortions experienced by households in this context. It may not have fully alleviated credit constraints: given that average monthly savings were US$8.9 and the loan had a four-month repayment period, someone with a binding savings constraint might only be able to pay US$36 over the period of the loan. We therefore estimate the stylized model specified in Section 2 and generate counterfactuals to quantify outcomes in alternative scenarios where the demand distortion is alleviated further. Of course, efficient WTP in this context may not be exactly equal to the discounted stream of fuel savings: other factors like risk aversion or higher time discounting could also lower WTP relative to the total stream of savings causing $\gamma > 0$, or the induction stove could generate additional unobservable benefits such that WTP exceeded savings. Still, we think the full stream of financial benefits discounted at 10% per year is a useful benchmark against which to compare the results.

The model clarifies how the demand distortion and the private-social benefit correlation affect subsidy efficacy. We are agnostic about what drives this heterogeneity. For example, in another context a clean technology might generate a reduction in pollution exposure that increases WTP even in the absence of any financial savings: this would still cause a positive correlation between private and social benefits, and the model's insights would still apply.

As discussed in Section 5.1 and Section 6.3, the results are consistent with most of the wedge being an allocation distortion. For simplicity, this section therefore assumes an allocation distortion throughout, affecting the agent's decision-making without having any direct welfare consequences.

7.1 Set-up

We assume agents i are distributed Weibull by their private benefit from adopting: $(b_i/\hat{b}) \sim \text{Weibull}(\Omega_1, \Omega_2)$, with discounted private benefits after 2.2 years $\hat{b} = \text{US\$209}$ (from panel a, column 7 of Table 1). The Weibull's shape parameter governs tail thickness, allowing the data to discipline how concentrated WTP is around its mean; for $\Omega_1 \geq 1$ it satisfies the assumed increasing hazard rate (IHR) condition (the estimate of $\Omega_1 = 1.9$ satisfies this condition). The correlation between social benefit and private benefit is controlled by Ω_3, as follows: $\phi_i = (\hat{\phi}/\hat{b})\,[\Omega_3\, b_i + (1-\Omega_3)\,\varepsilon_i]$, with $\hat{\phi} = 5.5$ tCO_2e (from panel b, column 4 of Table 1 multiplied by 2.2 years), $\Omega_3 \in [0,1]$, and ϵ_i an independent draw from the same Weibull distribution.

For each credit condition $c \in \{0,1\}$ the mean demand distortion $\gamma_c \equiv E[\gamma_i|c]$ is defined by $w\bar{t}p_c = (1-\gamma_c)\hat{b}$, with observed mean WTP $w\bar{t}p_0 = \text{US\$20}$ and $w\bar{t}p_1 = \text{US\$35}$ (from Table A4), giving $\gamma_0 = 0.91$ and $\gamma_1 = 0.83$ (from Table 3). We assume a heterogeneous demand distortion $\gamma_i = \text{logistic}(\mu_c + \sigma_c\eta_i)$, $\eta_i \sim N(0,1)$, with dispersion $\sigma_c = \log(1+\exp(\Omega_4 + \Omega_5(1-\gamma_c)))$, such that Ω_4 governs baseline dispersion and Ω_5 controls how dispersion varies with the mean wedge, allowing for heterogeneity in credit impacts to affect the dispersion of WTP.[21] The location parameter μ_c is pinned down by $E[\gamma_i|c] = \gamma_c$. Individual WTP is then given by $wtp_i = (1-\gamma_i)\, b_i$.

The distributions of wtp_i and ϕ_i can be used to calculate demand elasticity ϵ^D and the average externality of adopters who are marginal to the subsidy ($\bar{\phi} = \mathbb{E}[\phi_i | p - s_h \leq wtp_i < p - s_l]$, where $p - s_h = \text{US\$20}$ and $p - s_l = \text{US\$73}$ as in the experiment).

7.2 Model estimation

We target seven empirical moments: extensive margin elasticity ϵ_c^D, average marginal externality $\bar{\phi}_c$, and cost per marginal adoption—all of which vary by credit groups $c = \{0,1\}$—as well as a common $\nu = corr(b_i, \phi_i)$. For each moment n we define the simulated moment error $e_n(\mathbf{\Omega})$ as a normalized function of the empirical moment $m_n(x)$ and the simulated moment $\hat{m}_n(\tilde{x})$:

$$e_n(\mathbf{\Omega}) = \frac{\hat{m}_n(\tilde{x}) - m_n(x)}{m_n(x)}$$

[21]The logit-normal distribution across different $E[\gamma]$ reasonably approximates the LASSO-generated empirical distributions (Figure B4). Even though it also operates on the interval [0,1], we do not apply the Beta distribution because its behavior at the boundaries of [0,1] is inconsistent with what we observe in our data.

We estimate the model parameters $\mathbf{\Omega} = \{\Omega_1, \Omega_2, \Omega_3, \Omega_4, \Omega_5\}$ using two-step generalized method of moments (GMM) with simulated moments, defining the estimator:

$$\hat{\mathbf{\Gamma}}_{GMM} = \mathbf{\Omega} \;:\quad \min_{\mathbf{\Omega}} \boldsymbol{e}(\mathbf{\Omega})^T \mathbf{W} \boldsymbol{e}(\mathbf{\Omega})$$

We simulate 100,000 agents per parameter evaluation, with a fixed seed for reproducibility. The first-step weighting matrix is $\mathbf{W}_1 = \mathbf{I}$; the optimal second-step weight is $W = \hat{\mathbf{\Gamma}}^{-1}$, where $\hat{\mathbf{\Gamma}}$ is the sample covariance of the moment-error vector across 30 bootstrap re-simulations at the first-step estimate. The reported estimate corresponds to the lowest objective value at convergence across multiple starting values.

The estimated coefficients are $\hat{\Omega}_1 = 1.9$, $\hat{\Omega}_2 = 1$, $\hat{\Omega}_3 = 0.6$, $\hat{\Omega}_4 = -2.7$, and $\hat{\Omega}_5 = 31$. Panel a of Table 6 compares the moments simulated with the estimated parameters and the empirical moments calculated using the data. The subsidy costs per tCO_2e implied by the targeted moments, US\$11 and US\$20, are close to the empirically estimated values of US\$11 and US\$19, respectively. With these estimates in hand, we run counterfactuals by simulating $b_i \sim \text{Weibull}(\hat{\Omega}_1, \hat{\Omega}_2)$ and $\epsilon_i \sim \text{Weibull}(\hat{\Omega}_1, \hat{\Omega}_2)$.

Counterfactual I: Demand distortion Panel b of Table 6 shows the results implied by the model for different assumptions of γ. Demand distortions that dampen demand increase the positive externality generated by the marginal adopter and demand elasticity, lowering the subsidy cost per tCO_2e. In the neoclassical case where private benefits equal WTP—corresponding to no demand distortion—the subsidy cost per tCO_2e increases to US\$162.

We estimate two alternative specifications of the model to test the robustness of these results to alternate distributions of b_i and γ_i. The first assumes linear distributions, with benefits following a downward-sloping distribution over $[0,\Omega_1]$ and the distortion distributed over [0,1] with a triangular distribution parameterized by its mode, which yields three parameters. The second uses the same distribution of benefits as the main specification but estimates a logit-t distribution instead of a logit-normal distribution. Both simulations show worse moment fit but similar patterns across all four measures of subsidy efficiency (Table B9 and Table B10).

Counterfactual II: The correlation between WTP and social benefits We extend the counterfactuals to include alternative correlations between WTP and social benefits. Figure 8 shows the results when varying not only the demand distortion parameter γ but also the correlation parameter Ω_3. To help build intuition, we use labels E'_+, E_+, E'_-, and E_- to demarcate the approximate externalities (also labeled in panels a and b of Figure A1).

When WTP reflects private benefits and there is no distortion ($\gamma = 0$, such that $wtp_i = b_i$), a positive correlation induces adverse selection, decreasing the marginal positive externality ($E_+ < E_-$). On the other hand, under a demand distortion that dampens WTP ($\gamma > 0$), a positive correlation means compliers generate larger positive externalities, inducing beneficial selection ($E'_+ > E'_-$). The effect of a demand distortion thus hinges on the correlation between WTP and social benefits:

Table 6: Simulated estimates from observed and counterfactual parameters

(a) Estimation

Moment	RCT: Pay up front ($\gamma = 0.91$)			RCT: Pay with loan ($\gamma = 0.83$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	0%	0.8	0.8	0%
Marginal externality (ϕ)	5.7	5.8	1%	4.8	4.9	1%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.4	-1%	-0.9	-0.7	18%
USD per marginal adoption	65.4	64.7	-1%	93.4	97.9	5%

(b) Counterfactual simulations: Varying the distortion γ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	7.8	5.8	4.9	4.5	3.8	3.3	2.8
Extensive margin elasticity ($\mathcal{E}_D$)	-8.9	-2.4	-0.7	-0.4	-0.2	-0.1	-0.1
USD per marginal adoption	62	65	98	143	275	374	458
Observable welfare gain per subsidy dollar (in USD)							
Observed		13	7				
Simulated	20	13	6.9	4.4	1.8	1	0.7
Subsidy cost per tCO_2e (in USD)							
Observed		11	19				
Simulated	7.9	11	20	32	73	114	162

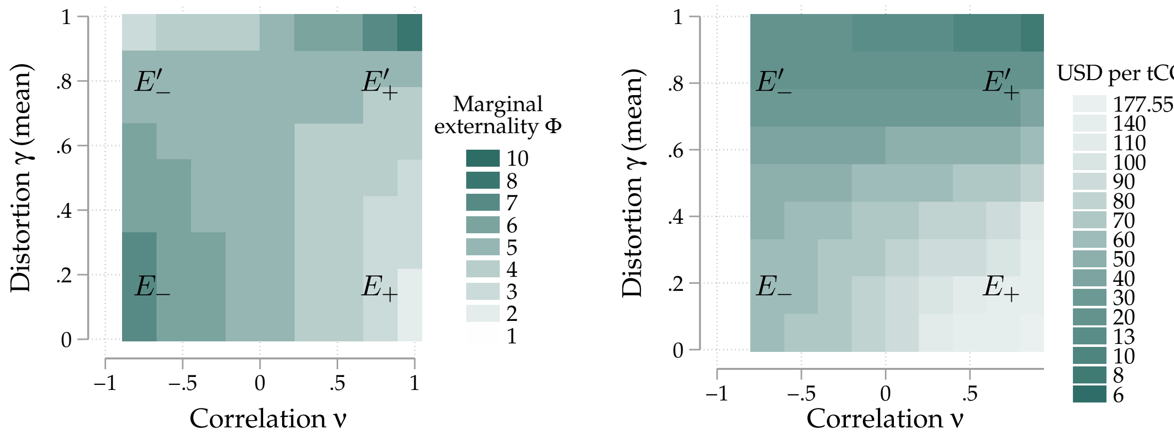
Notes: Panel a compares empirical moments and the simulated moments under the estimated model parameters. There are seven moments: a single correlation moment ν, and one per treatment group for each of the other three moments. Panel b uses the model parameters to estimate key moments, as well as the aggregate subsidy cost per tCO_2e, for counterfactual scenarios with higher or lower demand distortions, holding constant the correlation between private benefits and social benefits to match the empirical moment and assuming $r = 0$. Table B8 reports the same estimates with standard errors and confidence intervals. Table B7 shows estimates for the Marginal Value of Public Funds (MVPF) and resource cost per ton. Table B6 also varies the correlation parameter ν. Table B9 and Table B10 show that these results are not sensitive to the specific distributional assumption of b_i and γ_i.

when this correlation is negative, a demand distortion decreases the average externality of compliers ($E'_- < E_-$). It is only when it is positive that the distortion increases the average externality of compliers ($E'_+ > E_+$).

Panel b shows how both channels together affect the subsidy cost per tCO_2e. While the marginal externality channel depends on whether the correlation is positive or negative, the demand distortion increases the extensive margin elasticity, regardless of the correlation between WTP and social benefits. In this context, this latter effect outweighs the opposing effect on the average externality of compliers among negative correlation coefficients: even though E'_- has a lower marginal externality than E_- (panel a), it still has a lower subsidy cost per tCO_2e (panel b). As a result, the demand distortion unambiguously decreases the subsidy cost per tCO_2e in this context, regardless of the correlation coefficient ($E'_- < E_-$ and also $E'_+ < E_+$).

Figure 8: Counterfactual estimates

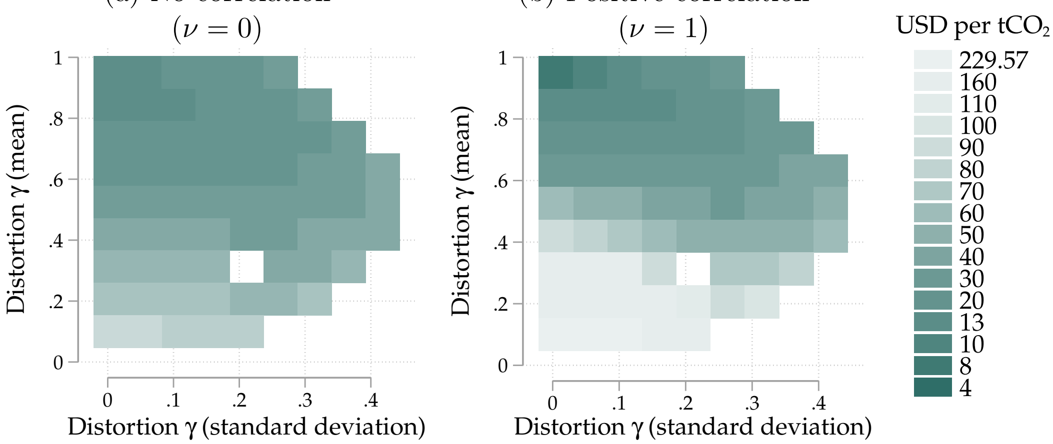
(a) Marginal externality (tCO_2e) (b) Subsidy cost per tCO_2e (US$)



Notes: Both figures show how demand distortions as well as the correlation between WTP and social benefits affect subsidy efficacy. Panel a shows the marginal externality. The labels E'_+, E_+, E'_-, and E_- correspond to the labels in panels a and b of Figure A1, which presents the theoretical underpinnings for channels 1 and 2. Panel b shows the combined effect of the marginal externality and the elasticity on the subsidy cost per tCO_2e. The demand distortion decreases the subsidy cost per tCO_2e everywhere in this context, regardless of correlation. Table B6 presents a subset of these estimates in the format of Table 6.

Figure 9: Counterfactual estimates with heterogeneous demand distortions

(a) No correlation ($\nu = 0$) (b) Positive correlation ($\nu = 1$)



Notes: This graph visualizes how different demand distortions and correlations affect the subsidy cost per tCO_2e when demand distortions are allowed to be orthogonally heterogeneous across agents. Since we constrain $\gamma \in [0, 1]$, not all combinations of mean and standard deviation are feasible.

Counterfactual III: Heterogeneous demand distortions across agents Finally, we explore how heterogeneity in the demand distortion affects subsidy efficacy. For simplicity we assume the γ_i are uncorrelated with any of the other parameters of interest (WTP and social benefits). We conduct the same counterfactual exercises as above, holding the correlation fixed at -1, 0, or $+1$. We draw $\gamma_i = \text{logistic}(\mu + \zeta\eta_i)$, $\eta_i \sim N(0, 1)$, varying μ and ζ to vary means and standard deviations of γ_i while ensuring $\gamma_i \in [0, 1]$.

Figure 9 shows the results. When the standard deviation is 0, the graphs follow the patterns in panel b of Figure 8: the mean distortion matters more when private and social benefits are positively correlated. Both panels show that this pattern weakens as the standard deviation increases.

8 Conclusion

Demand distortions such as financial market frictions and behavioral biases are pervasive across many low- and middle-income countries, often operating by dampening willingness-to-pay (WTP) for profitable technologies. This paper shows that this has important implications for the efficiency of Pigouvian price instruments. In theory, subsidies for green technologies generate a larger gain in observable welfare per subsidy dollar as well as a larger positive externality when agents face demand distortions that dampen demand. This operates through two channels: increasing the private and social benefits of the marginal adopter, and increasing demand elasticity.

We test this theory in a field experiment with more than 2,000 households in Nakuru County, Kenya, where we randomize marginal-cost subsidies, fixed-cost subsidies, and demand distortions for the purchase of an energy efficient technology: induction stoves. We first use the random variation in prices to estimate the causal impacts of stove adoption. We find that the stove abates 5.5 tCO_2e and reduces (discounted) household energy expenditures by US$209 over the 2.2 years of expected usage of the stove. WTP among the control group is US$20: access to a loan increases WTP to US$35. Even factoring in continued use of charcoal stoves and imperfect additionality of the subsidies, green subsidies in this context offer a very low-cost abatement opportunity, at only US$11 per ton of CO_2e abated. This compares favorably to many other abatement technologies available today. Factoring in the private benefit to adopters, the externality benefit, and the stove manufacturing cost, each dollar of subsidy generates US$12.6 in observable welfare benefit.

As predicted by the theory, demand distortions in this context contribute to these favorable numbers. The randomly assigned access to credit, which lowers the demand distortions, lowers the observable welfare benefit per subsidy dollar to US$7.2 and raises the subsidy cost per tCO_2e to US$19. This operates through the two channels identified in the model: demand distortions increase the marginal adopter's positive externality by 19% and increase demand elasticity from -0.9 to -2.3.

The loan treatment increases WTP in this context but still does not close the adoption gap. To estimate the subsidy cost per tCO_2e in the absence of any demand distortions, we estimate the model using two-step generalized method of moments (GMM), using the random variation from the experiment to identify key parameters. Counterfactuals suggest that if WTP equaled the discounted stream of fuel savings, the observable welfare gain per subsidy dollar would have fallen by a factor of 19, from US$12.6 to US$0.7, and the subsidy cost per tCO_2e would have increased more than tenfold, to US$162.

The mechanisms identified in this paper suggest that many of the most effective abatement opportunities may be found in contexts with larger demand distortions. As an example, the U.S. has relatively sophisticated financial markets for consumer durables, with the vast majority of new cars sold with financing. Our paper suggests that this could contribute to the high subsidy cost per tCO_2e of electric vehicle subsidies, at more than US$1,200.

Approximately 1.4 million Kenyan households use charcoal on a daily basis. Optimistically, a price subsidy could incentivize adoption by up to 37% of households.[22] Peak-hour usage at the 90^{th} percentile in our sample is 88 Wh per customer, so scaling our experimental results would require an additional 45 MW in grid capacity, against Kenya's current peak demand of 2,177 MW and installed capacity of 3,243 MW (Kenya Power, 2024). Kenya's electric utility appears capable of meeting the demand increase associated with widespread electric stove adoption.

Approximately 54 million households across the world today use charcoal as their primary cooking fuel, including approximately 31 million households that live in Africa (Berkouwer and Dean (2026c)). At annual abatement of 2.6 tCO_2e and with 37% adoption at subsidized prices, a widespread cooking transition in Africa would abate approximately 30 million tons of CO_2e per year. In addition, more than 300 million households across the world still cook primarily with firewood, and the IEA (2025) estimates that the number of households cooking with wood or charcoal in Africa in 2050 will, if anything, be higher than today.

More than US$1 trillion is invested towards climate change mitigation each year, much of which is targeted at rapid electrification combined with grid decarbonization. Maximizing the impact of these expenditures requires allocating this funding towards abatement opportunities at the lowest portion of the abatement cost curve. Given the results of this study, many of these opportunities may be found in low- and middle-income countries, where demand distortions are pervasive. Additional research is needed to identify specific mitigation opportunities, and to understand how the heterogeneity of demand distortions across countries affects global cooking transition pathways.

[22] This assumes universal electricity access among charcoal users, which is not completely unreasonable. Charcoal is used primarily in urban and peri-urban areas and urban electricity access was 91% in 2024 (KNBS, 2024).

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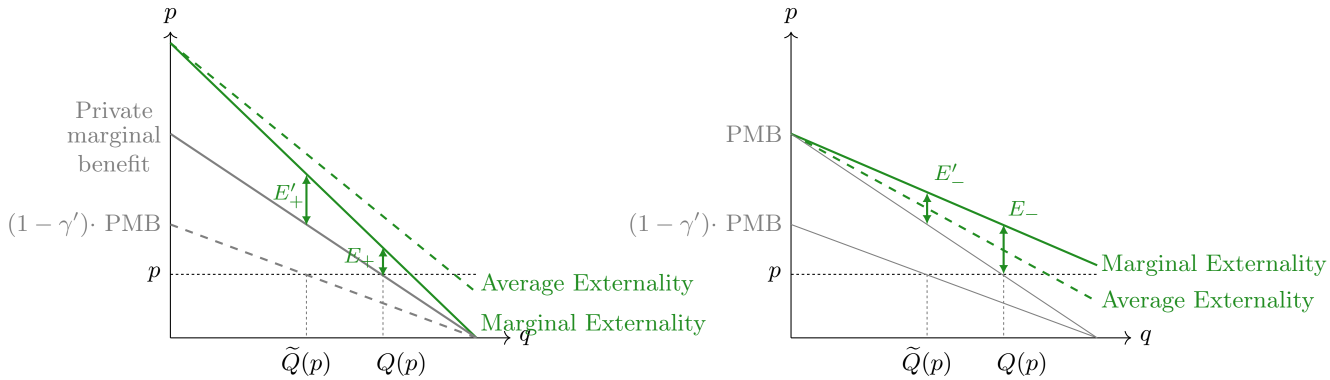
I Additional figures

Figure A1: How the shape of the externality determines the role of selection

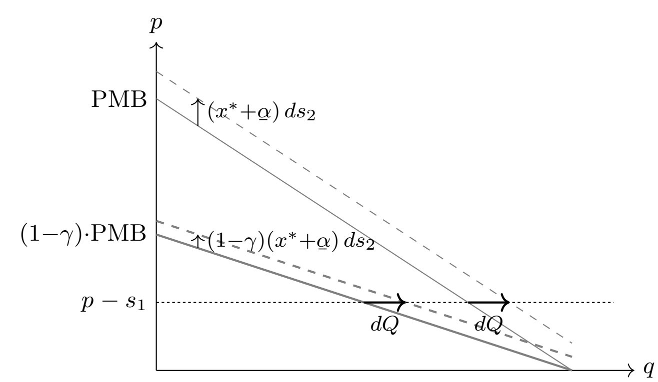
(a) Positive correlation
($\nu > 0$)

(b) Negative correlation
($\nu < 0$)



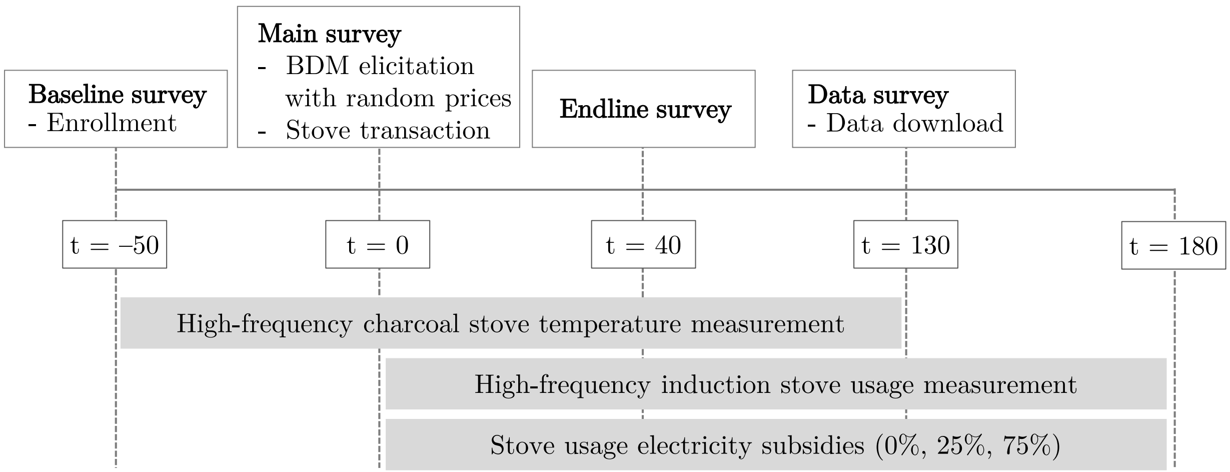
Notes: Demand distortions increase the marginal positive externality if willingness-to-pay and social benefits are positively correlated ($E'_+ > E_+$) but decrease it when they are negatively correlated ($E'_- < E_-$). A positive correlation generates advantageous selection that lowers the marginal externality of a subsidy in regular markets ($E_+ < E_-$) but increases it when markets face distorted demand ($E'_+ > E'_-$).

Figure A2: The distortion does not change the adoption response to the marginal-cost subsidy



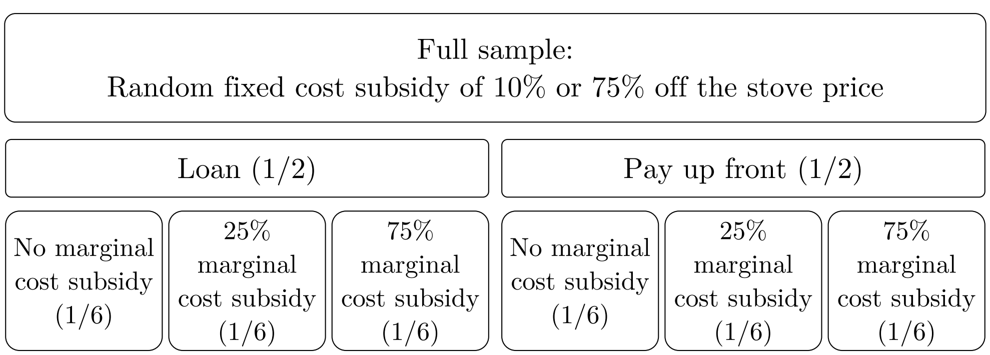
Notes: The distortion rotates perceived demand (PMB) around its quantity intercept to $(1-\gamma)\cdot$PMB (drawn for $\gamma = \frac{1}{2}$), making demand more price-responsive by the factor $1/(1-\gamma)$—the elasticity channel. A dollar of s_2 shifts each curve up by the marginal adopter's usage, damped to $(1-\gamma)(x^* + \underline{\alpha})$ on the rotated curve (dashed lines). The quantity gain at the price line $p - s_1$ equals the vertical shift divided by the slope, and both are scaled by the same factor $(1-\gamma)$, so the two horizontal arrows have equal length: adoptions per dollar of s_2 do not depend on γ except through the cutoff. To help build intuition, this graphical repesentation holds the marginal adopter fixed across the two demand curves, so the vertical shift is evaluated at a common $(x^* + \underline{\alpha})$ and the two quantity gains are exactly equal. This isolates the direct effect of s_2; in the model the cutoff itself moves with γ, and adoptions per dollar, $g(\underline{\alpha})(x^* + \underline{\alpha})/\pi'$, depend on γ only through that cutoff.

Figure A3: Experiment timeline



Notes: Timeline of study activities. The baseline survey was conducted on average 50 days before the main survey (10^{th}, 90^{th}: 40, 60 days), between August 12 and September 19, 2025. The main survey was conducted between 23 September and 19 November, 2025. The endline survey was conducted on average 40 days after the main survey ([25, 55]), between November 10 and December 12, 2025. The data download was conducted on average 130 days after the main survey ([110, 150]), between February 18 and March 6, 2026. High frequency induction stove usage monitoring and electricity subsidy transfers were completed remotely.

Figure A4: Randomized treatments



Notes: Random assignment for the 2,134 study participants. To preserve incentive-compatibility, 4% of the sample received a different subsidy between 10% and 75%. Section 3.3 discusses the randomized treatments in more detail.

Figure A5: Randomized information sheets

(a) Charcoal sheet

i. Information control

ii. Information treatment

(b) Electricity sheet

i. Information control

ii. Information treatment, marginal-cost subsidy control

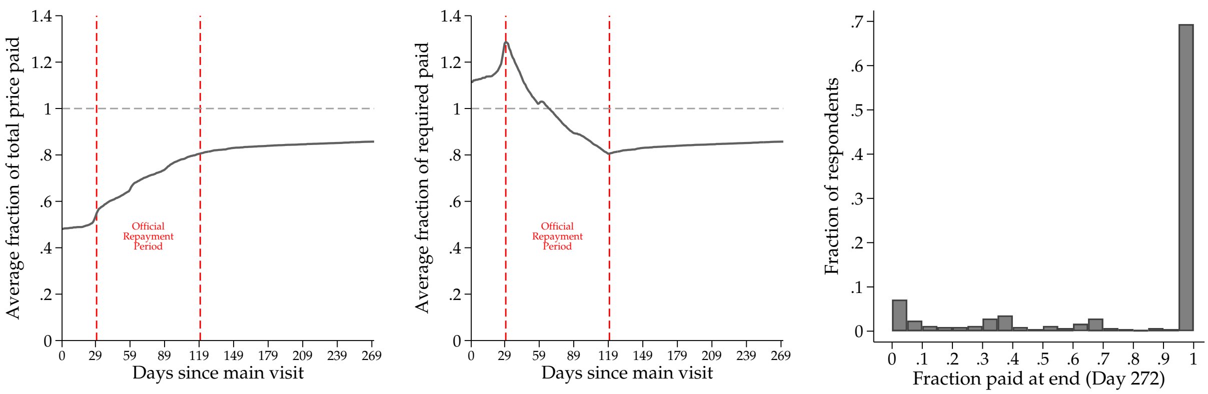
iii. Information treatment, 25% marginal-cost subsidy

iv. Information treatment, 75% marginal-cost subsidy

Notes: Each participant receives one charcoal sheet and one electricity sheet. 14% of participants are in the information control group and receive sheets with no cooking cost information (panel *i* in both panels). 86% of participants are in the information treatment group and receive cost information; respondents in the marginal-cost subsidy and information treatment groups saw cooking costs that incorporated the price reduction (panel *ii* in panel a and panel *ii*, *iii*, or *iv* in panel b).

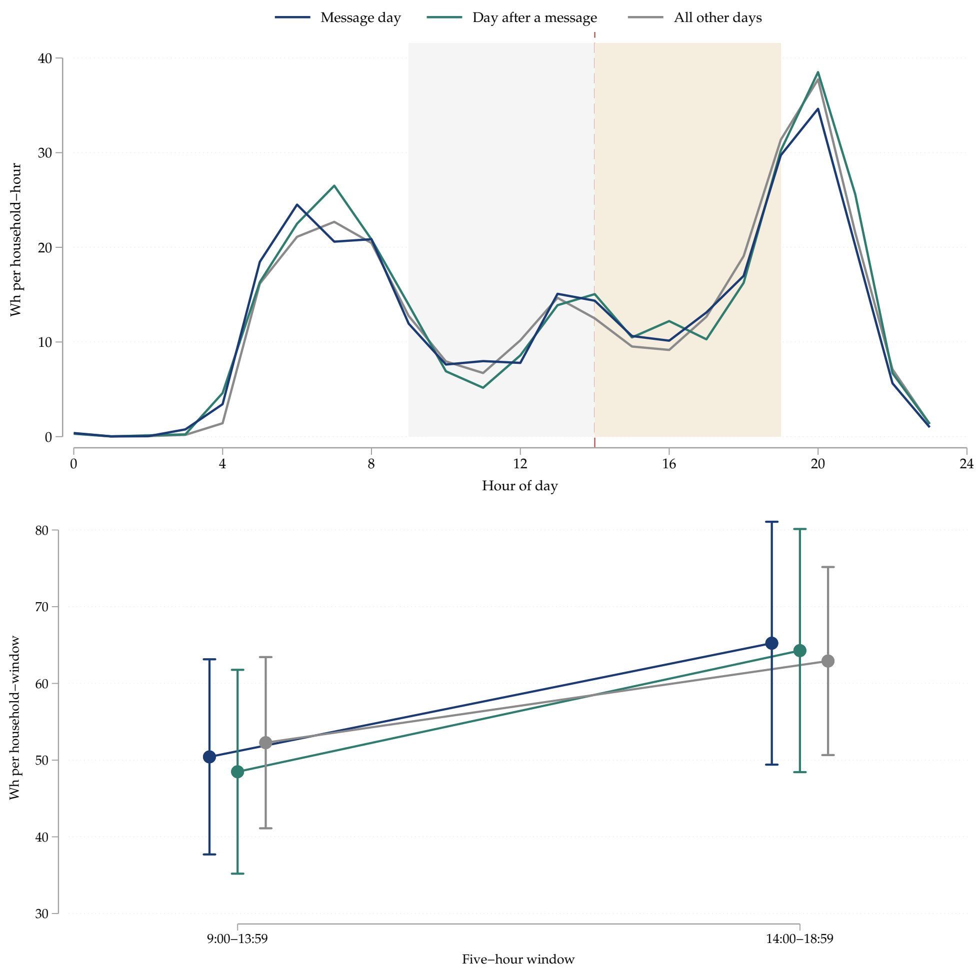
Figure A6: Loan repayment patterns

(a) Average cumulative fraction of price repaid over time (b) Average cumulative fraction of payplan repaid over time (c) Total fraction of price repaid at end



Notes: Repayment data for 271 days since the main visit, the longest common follow-up horizon across all respondents. The US$12 down payment required at day 0 constituted 16% of the price for those in the high price (low subsidy) group and 57% of the price for those in the low price (high subsidy) group. Panel a displays the average ratio of the amount respondents had paid to their total randomly assigned price on each day. In our study, borrowers have a 30-day grace period before the start of the official repayment period, which we mark as days 30–120 in the figure (dashed red lines). Panel b displays the average ratio of the amount respondents had paid to the amount cumulatively due on each day; values above 1 indicate respondents who were ahead of the smooth repayment schedule. Panel c displays a histogram of the fraction repaid by the 421 respondents in the credit treatments.

Figure A7: Results of the SMS experiment



Notes: Table B5 presents empirical estimates. The magnitudes are all small and none are statistically different from zero at the 10% level.

II Additional tables

Table A1: Summary statistics from respondent surveys at baseline

	Mean	SD	25^{th}	50^{th}	75^{th}	N
Number of residents	5.0	1.7	4	5	6	2134
Age	35.8	10.1	28	35	42	2134
Male (=1)	0.1	0.3	0	0	0	2134
Charcoal cooking time (minutes/day)	186.0	71.1	140	180	220	2134
Daily charcoal usage (KG per day)	3.1	1.0	2	3	4	2134
Number of charcoal stoves	1.1	0.3	1	1	1	2134
Charcoal spending (USD/month)	19.4	6.2	14	16	23	2134
Annual emissions from charcoal usage (tCO_2e)	4.9	1.5	3	4	6	2134
Number of LPG stoves	0.6	0.6	0	1	1	2126
LPG spending (USD/month)	9.1	12.0	8	8	9	1228
Annual emissions from LPG usage (tCO_2e)	0.3	0.4	0	0	0	1228
Number of wood stoves	0.1	0.4	0	0	0	2134
Wood spending (USD/month)	4.3	3.1	2	3	6	301
Respondent income (USD/month)	101.5	115.3	46	69	116	2127
Household income (USD/month)	246.9	204.8	116	196	300	1815
Monthly rent paid by household (30d; USD)	24.4	16.6	15	23	31	1797
Max willing to pay if phone broke (USD)	32.9	38.9	8	12	46	2134

Notes: Mean, standard deviation, and 25^{th}, 50^{th}, and 75^{th} percentiles for key socioeconomic variables collected during the baseline survey. 31% of participants who use wood report getting some (or all) of their monthly wood consumption for free; rather than noisily estimate emissions, we omit CO_2e emissions from direct firewood burning from our calculations (if anything, this likely underestimates total CO_2e abatement from electric stove adoption and understates aggregate subsidy efficiency).

Table A2: Effect of induction stove adoption on stove usage

	Ordinary Least Squares		Instrumental Variables				Joint cooking (IV)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7) Charcoal	(8) Electric	(9) Net
Pre-period difference	0.09		-0.15		-0.15				
	(0.08)		(0.18)		(0.18)				
Post-period difference	-0.43***		-0.52***		-0.52***		-0.45**	0.61***	0.14
	(0.07)		(0.16)		(0.16)		(0.18)	(0.04)	(0.18)
Pre-post difference		-0.49***		-0.38***		-0.38***			
		(0.06)		(0.13)		(0.13)			
Observations	220905	220905	5301720	5301720	220905	220905	199902	216827	199902
Respondents	2010	2010	2010	2010	2010	2010	1961	2080	1961
Control Mean	1.72	1.72	1.52	1.52	1.52	1.52	1.48	0.01	1.49
Level of data	Daily	Daily	Hourly	Hourly	Daily	Daily	Daily	Daily	Daily
Respondent FE?	No	Yes	No	Yes	No	Yes	No	No	No

Notes: Odd columns estimate the treatment-control difference in the pre-period (a baseline balance test) and in the post-period (a treatment effect estimate). Even columns estimate the treatment-control differences in the post period relative to the treatment-control difference in the pre-period, where the latter is absorbed by respondent fixed effects. All regressions cluster standard errors by respondent. All regressions include date and days-since-main-visit fixed effects. Columns 3 and 4 use hourly data and include hour-of-day fixed effects; the remaining columns use daily data. Column 3 uses hourly charcoal stove usage data and most closely represents a weighted average of the pre-period and post-period treatment effects shown in Figure 4; it replicates the specification reported in column 1 of Table 1. The 'post-period' coefficient in column 5 is our preferred specification for the causal impact of stove adoption on daily cooking time, and gives the same estimate.

Table A3: Alternative assumptions

	Assumption	Subsidy cost per tCO_2e
	Benchmark	US$11
(1)	One-year durability	US$25
(2)	Three-year durability	US$8
(3)	Fraction non-renewable biomass (fNRB) of 10%	US$14
(4)	Fraction non-renewable biomass (fNRB) of 100%	US$8
(5)	US grid emissions intensity	US$12

Notes: Subsidy efficiency results under alternative assumptions. The benchmark is this paper's main estimate. Scenarios (1) and (2) assume that the stove's expected life span is one year or three years (the main estimate uses 2.2). Scenarios (3) and (4) assume that the fraction of the wood used for charcoal production that is non-renewable is either 10% or 100% (the main estimate assumes 38%). Scenario (5) assumes that Kenya's average grid emissions were those of the U.S., at 366 grams of CO_2e per kWh (the main estimate uses Kenya's grid average of 154 grams of CO_2e per kWh).

Table A4: Effect of credit and marginal-cost subsidy on willingness-to-pay

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Credit treatment	14.1***	14.2***	14.7***	15.8***	14.1***	14.2***	14.7***	15.8***
	(2.1)	(2.1)	(1.9)	(2.0)	(2.1)	(2.1)	(1.9)	(2.0)
Subsidy	-2.0	-2.1	-1.5	-1.5				
	(1.3)	(1.3)	(1.2)	(1.3)				
Subsidy=25					-2.2	-2.3	-0.8	-0.4
					(1.5)	(1.4)	(1.5)	(1.5)
Subsidy=75					-1.7	-1.8	-2.3	-2.5*
					(1.5)	(1.4)	(1.4)	(1.4)
Credit X Subsidy (any)	2.4	2.3	2.2	1.6				
	(2.6)	(2.5)	(2.4)	(2.4)				
Credit X 25% Subsidy					0.8	0.9	0.3	-0.6
					(3.0)	(2.9)	(2.8)	(2.9)
Credit X 75% Subsidy					4.0	3.8	4.0	3.7
					(2.9)	(2.9)	(2.7)	(2.7)
Observations	2134	2134	2134	2134	2134	2134	2134	2134
Control Mean	19.5	19.5	19.5	19.5	19.5	19.5	19.5	19.5
SES controls	None	Some	Many	All	None	Some	Many	All

Notes: The effect of the credit treatment, the electricity subsidy treatment, and their interaction on willingness-to-pay. Columns 1, 2, 3, and 4 (as well as 5, 6, 7, and 8 respectively) vary only in using either no, some, many, or all socioeconomic controls.

Table A5: Impact of credit on daily induction stove usage

	All buyers		Marginal buyers	
	(1) Daily	(2) Respondent	(3) Daily	(4) Respondent
Credit (=1)	-95.6***	-97.4***	-83.0**	-86.8**
	(27.4)	(30.0)	(33.8)	(39.8)
Observations	109994	625	62875	358
Respondents	625		358	
Control Mean	415	415	404	403

Notes: Results from ordinary least squares regressions estimating how credit constraints affect the composition of marginal adopters. Standard errors in parentheses are clustered by respondent. Regressions include date fixed effects and control for randomization strata and other treatments. Columns 1 and 3 use daily data, with standard errors clustered by respondent. Columns 2 and 4 use respondent daily average.

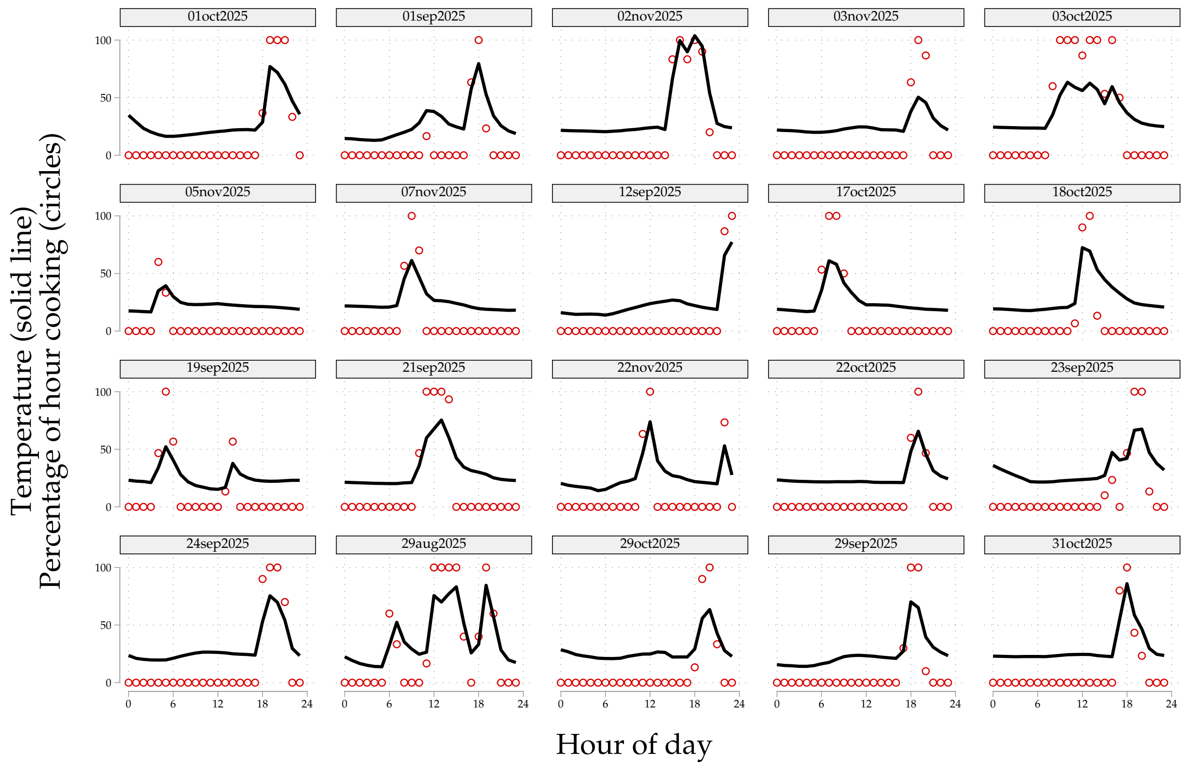
Table A6: Extensive and intensive margin impacts of electricity subsidy

	Willingness-to-pay			Daily induction stove usage		
	(1) USD	(2) USD	(3) USD	(4) Wh	(5) Wh	(6) Log(Wh)
Any subsidy	-0.41			17.29		
	(1.22)			(32.37)		
=25		-0.51			5.69	
		(1.44)			(38.14)	
=75		-0.31			29.77	
		(1.39)			(37.91)	
Subsidy value (USD)			-0.02			
			(0.12)			
Log(Marginal price)						-0.03
						(0.05)
Observations	2134	2134	2134	111599	111599	48904
Respondents				625	625	614
Control mean	28	28	28	421	421	

Notes: Results from ordinary least squares regressions estimating how the electricity subsidy affects outcomes. Columns 1 and 4 pool the two subsidy levels of 25% and 75%; the remaining columns separate the levels. All columns control for randomization strata and other treatments. Columns 4-6 use data at the respondent-by-day level with date fixed effects and standard errors in parentheses clustered by respondent (results are qualitatively indistinguishable when averaging daily usage to the respondent level).

III Online Appendix Figures and Tables

Figure B1: Temperature-to-cooking conversion algorithm examples



Notes: Comparison of raw temperature data in Celsius (black line) and the estimated cooking fraction (red dots), for 20 unique dates and unique respondents, randomly selected from the set with non-zero cooking events. Section IV.3 describes the technology and the conversion algorithm.

Table B1: Balance on main treatment variables

	Sample Mean	Credit Treatment	Low BDM Price (<30)	Electricity Treatment	N
	(1)	(2)	(3)	(4)	
Stratification variable: Charcoal spending	19.43	0.23	0.02	-0.04	2134
(USD/month)	[6.20]	(0.27)	(0.27)	(0.28)	
Stratification variable: Use lpg/gas/ethanol for	0.58	0.00	0.00	0.00	2134
cooking (=1)	[0.49]	(0.02)	(0.02)	(0.02)	
Stratification variable: Electricity top-up	4.68	0.39**	-0.11	-0.20	2134
spending (USD, 5 weeks)	[4.32]	(0.19)	(0.19)	(0.20)	
Household size	4.95	0.01	0.08	0.03	2134
	[1.65]	(0.07)	(0.07)	(0.08)	
Age	35.84	0.06	-0.26	-0.18	2134
	[10.15]	(0.44)	(0.44)	(0.47)	
Male respondent	0.10	0.00	0.02	0.01	2134
	[0.30]	(0.01)	(0.01)	(0.01)	
Charcoal cooking time	186.00	-2.02	-3.09	3.00	2134
(minutes/day)	[71.08]	(3.08)	(3.08)	(3.26)	
LPG/ethanol spending (USD/month)	5.23	0.54	-0.49	-0.06	2128
	[10.16]	(0.44)	(0.44)	(0.47)	
Respondent income (USD/month)	101.44	2.33	1.09	-2.86	2127
	[115.35]	(5.00)	(5.01)	(5.29)	
Household income (USD/month)	246.99	16.12*	16.38*	-12.88	1815
	[204.86]	(9.61)	(9.62)	(10.18)	
Blackout hours (last 30 days)	13.93	2.24*	0.50	-0.44	2134
	[27.49]	(1.19)	(1.19)	(1.26)	
Voltage fluctuation hours (last 30 days)	2.04	-0.70	-0.44	0.20	2134
	[10.88]	(0.47)	(0.47)	(0.50)	
Joint F-Test		0.09	0.09	0.21	

Notes: Test for balance of baseline variables. We only include participants who completed the main survey because they are the only participants whom we assigned to a treatment group.

Table B2: Attrition by treatment assignment and adoption status

	Control Mean	Attrited (Visit 3)
Credit treatment (=1)	0.030	-0.010
		(0.007)
BDM price < 30 USD	0.027	-0.005
		(0.007)
Electricity treatment > 0	0.021	0.006
		(0.007)
Bought induction stove (=1)	0.031	-0.022***
		(0.007)
Joint F-Test p-Value		0.00
Sample Mean		0.02

Notes: 2,134 respondents completed the main survey and 2,081 completed the endline survey. Predictors of attrition (defined as not completing an endline visit) among the sample of respondents who completed the main survey.

Table B3: Attrition: Participant availability

Reason	Frequency
Completed survey	2081
Unavailable	8
Withdrew from study	8
Relocated outside survey team reach	36
Hospitalized	1
Total	2134

Notes: Completion of the endline survey among the 2,134 respondents who completed the main survey. Participants were classified as relocated outside survey team reach if they had moved outside the survey area or outside Kenya either permanently or for at least 3 months and thus could not be contacted for follow-up.

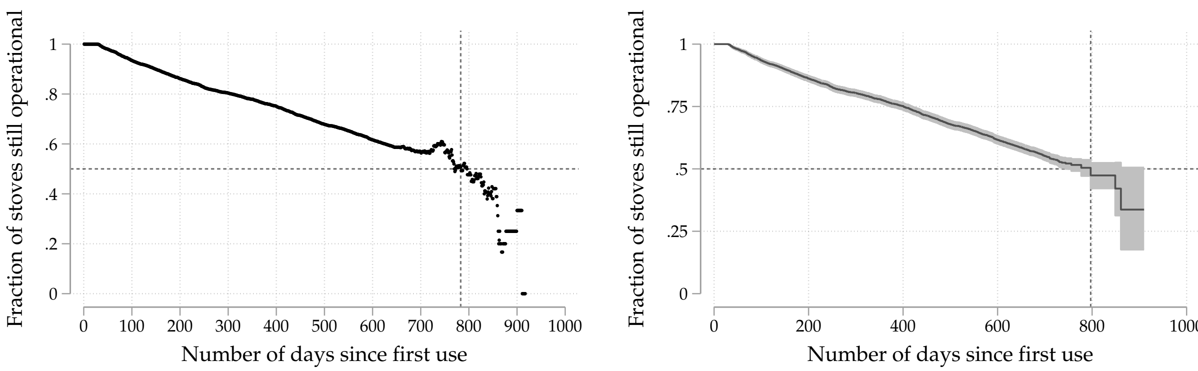
Table B4: Robustness of credit treatment effects to default

	(1) WTP	(2)	(3)	(4)
Credit	16.11***	6.77***	9.77***	8.84***
	(1.14)	(0.60)	(1.01)	(0.96)
Observations	2134	2134	2134	2134
Control Mean	19.51	4.31	19.51	19.51

Notes: Robustness of the effect of the credit treatment on willingness-to-pay, adjusting for observed repayment rates. Column 1 replicates the main effect of credit on WTP. Column 2 uses the amount of money paid, counting those who did not adopt as zero. Column 3 replaces willingness-to-pay with the amount paid for those who paid less than 100%. Column 4 deflates WTP in the credit treatment by the average rate of default.

Figure B2: Stove lifespan estimation

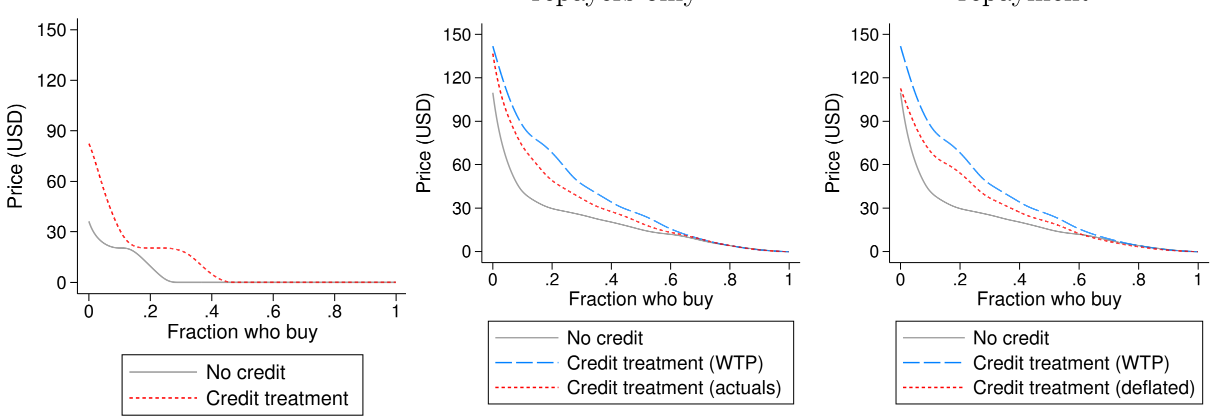
(a) Data (b) Estimated survival function



Notes: Stove usage rates for the 23,444 stoves that had been sold on or before December 31, 2024. The time variable is the number of days between first usage and last usage. An adverse event is the last observed usage if that usage happened more than 3 months before the end of our dataset. We are agnostic as to the cause of the adverse event: it likely includes breakage as well as non-payment with stove lockout. We drop the 2% of stoves with less than 24 hours of usage as well as the 3% of stoves with usage between 1 and 29 days since faulty stoves identified within this period are replaced for free by the seller. Median lifespans are marked by the dotted lines. Panel a shows the empirical distribution of survival. Panel b shows the survival curve estimated using a Cox proportional hazard model, according to which the median induction stove sold by the seller has a lifespan of 2.2 years.

Figure B3: Willingness-to-pay adjusting for repayment rates

(a) Amounts paid (b) Amount paid for partial repayers only (c) WTP times average repayment



Notes: Demand curves that adjust for observed repayment rates. Panel a presents the actual amounts paid by the treatment group, counting non-adopters as zeros. Panel b replaces willingness-to-pay (WTP) with the actual amount paid only for buyers with credit who did not pay 100% of the loan. Panel c deflates WTP across the distribution by the average default rate. Columns 2, 3, and 4 of Table B4 present regression estimates of the impact of the credit treatment using these three definitions, respectively.

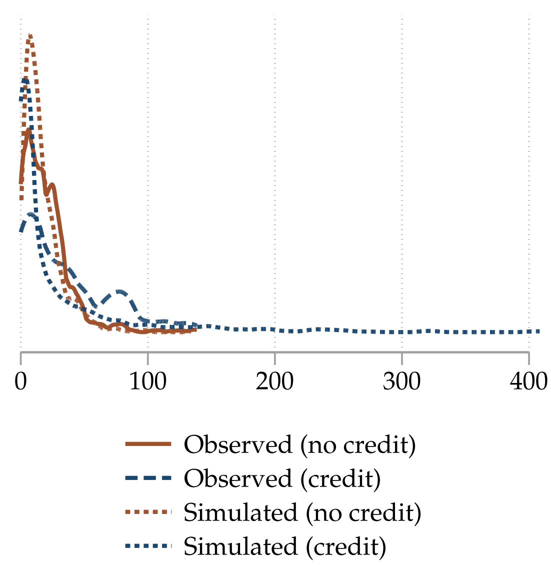
Table B5: Results of the SMS experiment

	Usage (kWh)		Log usage		Inverse hyperbolic sine	
	(1)	(2)	(3)	(4)	(5)	(6)
Message day	-0.024	-0.024	-0.014	-0.014	-0.014	-0.014
	(0.024)	(0.024)	(0.082)	(0.082)	(0.014)	(0.014)
× 25% subsidy	0.034		0.099		0.021	
	(0.037)		(0.096)		(0.023)	
× 75% subsidy	-0.005		-0.139		-0.003	
	(0.030)		(0.089)		(0.019)	
× any subsidy		0.014		-0.022		0.009
		(0.030)		(0.084)		(0.018)
Own week	0.015	0.015	0.063	0.063	0.007	0.007
	(0.016)	(0.016)	(0.050)	(0.050)	(0.010)	(0.010)
Observations	31,081	31,081	9,552	9,552	31,081	31,081
Households	578	578	400	400	578	578
Control Mean	0.351	0.351				

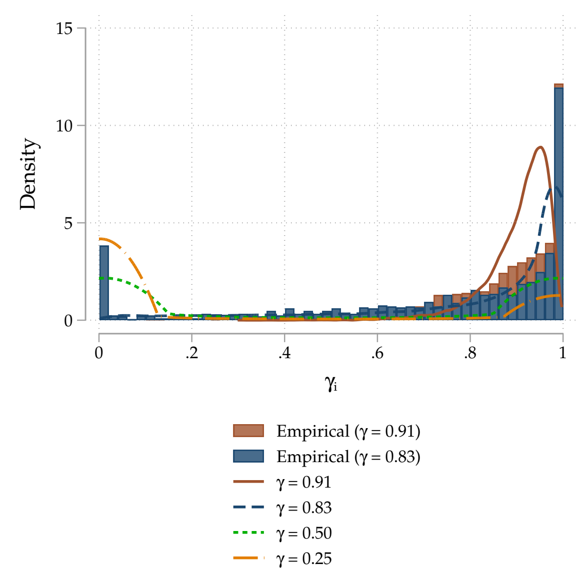
Notes: Standard errors clustered by household throughout.

Figure B4: Empirical and simulated WTP and gamma distributions

(a) Willingness-to-pay



(b) Demand distortion (γ_i)



Notes: Panel a: Simulated and observed distributions of WTP by credit treatment group. Panel b: The histograms show the empirical distributions of the γ_i after using LASSO to estimate heterogeneous treatment effects and then projecting resulting estimates onto respondent characteristics to predict their individual savings, and then dividing WTP by the result to estimate individual-level γ_i. The four lines show $\gamma_i = \text{logistic}(\mu_c + \sigma_c \varepsilon_i)$, $\varepsilon_i \sim N(0, 1)$, with dispersion $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma_c)))$ for the estimated values of Ω_4 and Ω_5 across four different values of γ_c. Even though GMM estimates 5 parameters to match 7 moments simultaneously, the distributions appear to reasonably capture the estimated underlying distribution of γ_i.

Table B6: Simulated estimates varying both the demand distortion and the correlation between private and social benefits

	Assumed demand distortion (γ)	Assumed correlation (ν)	Average marginal externality (ϕ)	Extensive margin elasticity ($\mathcal{E}_D$)	Subsidy cost per tCO_2e (US$)	
(1) *Counterfactual*	.91	-.8	3.7	-2.4	64.4	17.4
(2) *Counterfactual*	.91	-.6	4	-2.4	64.4	16.1
(3) *Counterfactual*	.91	-.4	4.2	-2.4	64.4	15.2
(4) *Counterfactual*	.91	-.2	4.5	-2.4	64.4	14.4
(5) *Counterfactual*	.91	0	4.8	-2.4	64.4	13.5
(6) *Counterfactual*	.91	.2	5.1	-2.4	64.4	12.7
(7) *Counterfactual*	.91	.4	5.3	-2.4	64.4	12.1
(8) *Counterfactual*	.91	.6	5.6	-2.4	64.4	11.6
(9) RCT: Pay up front	.91	.8	5.8	-2.4	64.4	11.1
(10) *Counterfactual*	.91	1	6.6	-2.4	64.4	9.8
(11) *Counterfactual*	.83	-.8	4.6	-.7	97.1	21.1
(12) *Counterfactual*	.83	-.6	4.7	-.7	97.1	20.9
(13) *Counterfactual*	.83	-.4	4.7	-.7	97.1	20.7
(14) *Counterfactual*	.83	-.2	4.7	-.7	97.1	20.5
(15) *Counterfactual*	.83	0	4.8	-.7	97.1	20.3
(16) *Counterfactual*	.83	.2	4.8	-.7	97.1	20.1
(17) *Counterfactual*	.83	.4	4.9	-.7	97.1	20
(18) *Counterfactual*	.83	.6	4.9	-.7	97.1	19.9
(19) RCT: Pay with loan	.83	.8	4.9	-.7	97.1	19.8
(20) *Counterfactual*	.83	1	5	-.7	97.1	19.4
(21) *Counterfactual*	.5	-.8	6.1	-.2	274.7	44.7
(22) *Counterfactual*	.5	-.6	5.8	-.2	274.7	47.5
(23) *Counterfactual*	.5	-.4	5.5	-.2	274.7	50
(24) *Counterfactual*	.5	-.2	5.2	-.2	274.7	52.9
(25) *Counterfactual*	.5	0	4.8	-.2	274.7	57.1
(26) *Counterfactual*	.5	.2	4.5	-.2	274.7	61.1
(27) *Counterfactual*	.5	.4	4.3	-.2	274.7	64.6
(28) *Counterfactual*	.5	.6	4	-.2	274.7	68.2
(29) *Counterfactual*	.5	.8	3.8	-.2	274.7	72.9
(30) *Counterfactual*	.5	1	3	-.2	274.7	92
(31) *Counterfactual*	0	-.8	7.5	-.1	458.4	61.2
(32) *Counterfactual*	0	-.6	6.8	-.1	458.4	67.7
(33) *Counterfactual*	0	-.4	6.2	-.1	458.4	74
(34) *Counterfactual*	0	-.2	5.6	-.1	458.4	82.2
(35) *Counterfactual*	0	0	4.8	-.1	458.4	95.3
(36) *Counterfactual*	0	.2	4.2	-.1	458.4	108.6
(37) *Counterfactual*	0	.4	3.8	-.1	458.4	122
(38) *Counterfactual*	0	.6	3.3	-.1	458.4	137.8
(39) *Counterfactual*	0	.8	2.8	-.1	458.4	161.6
(40) *Counterfactual*	0	1	1.4	-.1	458.4	339.2

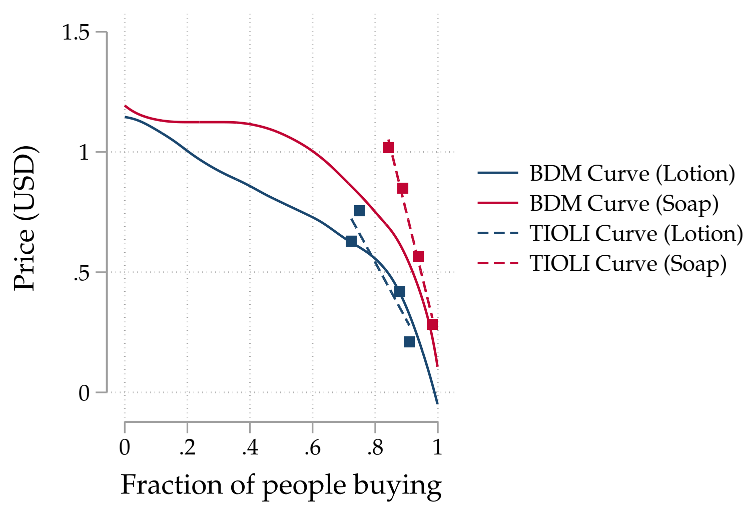
Notes: Additional counterfactual estimates using alternative values of demand distortion (γ) and correlation between private and social benefits (ν).

Table B7: Simulated marginal value of public funds and economic cost per ton from observed and counterfactual parameters

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	7.8	5.8	4.9	4.5	3.8	3.3	2.8
Extensive margin elasticity ($\mathcal{E}_D$)	-8.9	-2.4	-0.7	-0.4	-0.2	-0.1	-0.1
USD per marginal adoption	62	65	98	143	275	374	458
Resource cost per tCO$_2$e (in USD)							
Observed		-25	-19				
Simulated	-38	-25	-19	-18	-8	1	11
Marginal Value of Public Funds (in USD)							
Observed		14	8				
Simulated	21	14	8	5	3	2	2

Notes: Additional versions of panel b of Table 6. The economic cost of abatement is defined as the economic cost per adoption divided by the abatement per adoption: $\frac{p-\hat{b}}{\phi}$. Since fuel savings for the distorted groups exceed the US$82 production cost of the stove, the economic cost per ton is *negative*: each cookstove will *save* economic resources in this context. Abating one ton of CO$_2$*e* thus *reduces* total economic costs by US$25. Marginal Value of Public Funds (MVPF) uses the Hendren and Sprung-Keyser (2020) approach. Since willingness-to-pay is distorted and thus cannot be taken as a measure of value, we calculate this directly as the sum of cash transfer (valued at US$1) plus the stove adoption benefit per subsidy dollar (defined as the benefit per stove adoption divided by the subsidy cost per stove adoption): MVPF $= 1 + \frac{\hat{b}+\phi-p}{\mathbb{C}_s}$. This yields an MVPF of 3.2 when excluding environmental benefits and either 14, 20, or 37 when using a Social Cost of Carbon (SCC) of US$120, US$190, or US$380, respectively.

Figure B5: BDM and TIOLI elicitation results



Notes: Prior to the BDM each respondent completes two practice exercises, one for a bottle of lotion and one for a bar of soap. Each respondent is allocated a random price for the lotion and a random price for the soap. Respondents are randomly assigned whether they would be offered the lotion using take-it-or-leave-it ('TIOLI') and the bar of soap using BDM, or vice versa. These two practice take-up decisions serve two purposes. First, participants learn how the BDM mechanism works relative to the TIOLI they are used to in stores and experience the binding nature of the BDM. Second, comparing the demand curves elicited through the TIOLI and BDM mechanisms provides a test of comprehension. Their similarity for both goods suggests respondents understand the BDM mechanism and that the elicited WTP values reflect realistic decisions. The average price paid for these goods was US$0.68 (substantially less than the cost of the stove) and came out of the respondent's study compensation, and is therefore unlikely to affect the respondent's short-term liquidity or WTP directly.

Table B8: Simulated estimates with standard errors and confidence intervals

(a) Estimation

	RCT: Pay up front ($\gamma = 0.91$)			RCT: Pay with loan ($\gamma = 0.83$)		
Moment	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.80	0.80	0%	0.80	0.80	0%
s.e.	(0.010)	(0.001)		(0.010)	(0.001)	
Marginal externality (ϕ)	5.73	5.81	1%	4.82	4.89	1%
s.e.	(0.361)	(0.018)		(0.371)	(0.022)	
Extensive margin elasticity ($\mathcal{E}_D$)	-2.33	-2.36	-1%	-0.88	-0.72	18%
s.e.	(0.133)	(0.027)		(0.053)	(0.009)	
USD per marginal adoption	65.37	64.71	-1%	93.35	97.94	5%
s.e.	(0.739)	(0.096)		(2.891)	(0.600)	

(b) Counterfactual simulations: Varying the distortion γ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	7.8	5.8	4.9	4.5	3.8	3.3	2.8
	[7.1, 8.5]	[5.4, 6.3]	[4.5, 5.3]	[4.2, 4.9]	[3.5, 4.1]	[3.1, 3.5]	[2.7, 3.0]
Extensive margin elasticity ($\mathcal{E}_D$)	-8.9	-2.4	-0.7	-0.4	-0.2	-0.1	-0.1
	[-9.8, -7.2]	[-2.5, -2.2]	[-0.8, -0.6]	[-0.5, -0.3]	[-0.2, -0.2]	[-0.1, -0.1]	[-0.1, -0.1]
USD per marginal adoption	62	65	98	143	275	374	458
	[62, 62]	[64, 65]	[91, 105]	[128, 157]	[243, 310]	[324, 428]	[397, 539]
Observable welfare gain per subsidy dollar (in USD)							
Observed		13	7.2				
		[11, 14]	[6.3, 8.2]				
Simulated	20	13	6.9	4.4	1.8	1.0	0.7
	[18, 22]	[12, 14]	[6.2, 7.7]	[3.9, 4.9]	[1.6, 2.0]	[1.0, 1.1]	[0.6, 0.7]
Subsidy cost per tCO_2e (in USD)							
Observed		11	19				
		[10, 13]	[17, 22]				
Simulated	7.9	11	20	32	73	114	162
	[7.3, 8.7]	[10, 12]	[18, 22]	[28, 35]	[66, 79]	[105, 123]	[148, 178]

Notes: Table 6 with uncertainty attached. The point estimates are identical: this table does not re-estimate the model, it adds standard errors and intervals to the estimates reported there. Sampling uncertainty in the moments is obtained by resampling respondents and re-running each estimating equation; simulation uncertainty is the spread across simulation seeds at the estimated parameters. In panel a, the standard error below each empirical moment is the resampling standard error and the one below each simulated moment is the simulation standard error. In panel b, brackets are 95% intervals. For the simulated rows they are percentiles across 300 parameter vectors drawn from the estimated sampling distribution, with the counterfactual recomputed for each draw; for the two observed rows they come from the moment bootstrap directly. The interval on the subsidy cost per marginal adoption is degenerate in column 1 because at that level of distortion almost no agent has a willingness-to-pay above the low price, so the cost is pinned at s_h = US\$62 for every draw. The estimated parameters are $\hat{\Omega}_1 = 1.9$ (0.009), $\hat{\Omega}_2 = 1$ (0.038), $\hat{\Omega}_3 = 0.6$ (0.009), $\hat{\Omega}_4 = -2.7$ (0.285), and $\hat{\Omega}_5 = 31$ (2.96), all significant at the 1% level. The model is overidentified with 2 degrees of freedom; the Hansen J statistic is 18.1 ($p < 0.001$), so the overidentifying restrictions are rejected. The seven moments cannot be matched exactly by five parameters, which panel a shows directly: the extensive margin elasticity in the loan arm is the moment the model fits least well.

Table B9: Simulated estimates from observed and counterfactual parameters using alternate distributional assumption (1)

(a) Estimation

	RCT: Pay up front ($\gamma = 0.91$)			RCT: Pay with loan ($\gamma = 0.83$)		
Moment	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	0%	0.8	0.8	0%
Marginal externality (ϕ)	5.7	4.8	-16%	4.8	4.5	-7%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-0.6	74%	-0.9	-0.5	37%
USD per marginal adoption	65.4	108.3	66%	93.4	115.8	24%

(b) Counterfactual simulations: Varying the distortion γ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	4.8	4.8	4.5	4	3.6	3.6	2.8
Extensive margin elasticity ($\mathcal{E}_D$)	-0.6	-0.6	-0.5	-0.4	-0.3	-0.3	-0.2
USD per marginal adoption	108	108	116	137	178	178	289
Resource cost per tCO$_2$e (in USD)							
Observed		-25	-19				
Simulated	-20	-30	-21	-11	-5	-5	13
Observable welfare gain per subsidy dollar (in USD)							
Observed		13	7				
Simulated	6	7	5	4	3	3	1
Subsidy cost per tCO$_2$e (in USD)							
Observed		11	19				
Simulated	22	23	26	34	50	50	104
Marginal Value of Public Funds (in USD)							
Observed		14	8				
Simulated	7	8	6	5	4	4	2

Notes: Robustness check of Table 6 with alternative (linear) distributional assumptions. Benefits follow a decreasing triangular distribution over $[0,\Omega_1]$ with $f(b) = 2\frac{(\Omega_1 - b)}{\Omega_1^2}$. The distortion γ_i follows a triangular distribution on $[0, 1]$ with mean $\mu = \frac{2-\Omega_3}{3} + (\gamma - \gamma_0)$, where γ is the distortion assumed in the column heading and $\gamma_0 = 0.91$:

$$f(\gamma_i;\, \mu) = \begin{cases} \dfrac{2\gamma_i}{3\mu - 1} & 0 \leq \gamma_i \leq 3\mu - 1, \\ \dfrac{2(1-\gamma_i)}{2 - 3\mu} & 3\mu - 1 \leq \gamma_i \leq 1. \end{cases}$$

Panel a shows that the fit is worse. Panel b shows that the distributional assumption does not meaningfully affect the qualitative results from the counterfactual simulations.

Table B10: Simulated estimates from observed and counterfactual parameters using alternate distributional assumption (2)

(a) Estimation

Moment	RCT: Pay up front ($\gamma = 0.91$)			RCT: Pay with loan ($\gamma = 0.83$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	-1%	0.8	0.8	-1%
Marginal externality (ϕ)	5.7	5.7	0%	4.8	4.8	1%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.5	-6%	-0.9	-0.9	-7%
USD per marginal adoption	65.4	64.2	-2%	93.4	85.4	-9%

(b) Counterfactual simulations: Varying the distortion γ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	7.4	5.7	4.8	4.4	3.7	3.2	2.7
Extensive margin elasticity ($\mathcal{E}_D$)	-4.8	-2.5	-0.9	-0.5	-0.2	-0.2	-0.1
USD per marginal adoption	62	64	85	117	214	299	374
Resource cost per tCO_2e (in USD)							
Observed		-25	-19				
Simulated	-38	-25	-19	-19	-9	1	12
Observable welfare gain per subsidy dollar (in USD)							
Observed		13	7				
Simulated	19	13	8	5	2	1	1
Subsidy cost per tCO_2e (in USD)							
Observed		11	19				
Simulated	8	11	18	26	58	95	137
Marginal Value of Public Funds (in USD)							
Observed		14	8				
Simulated	20	14	9	6	3	2	2

Notes: Robustness check of Table 6 with an alternative distributional assumption: estimating the model with a heavy-tailed logit-t distribution such that $\gamma_i = \text{logistic}(\mu_c + \sigma_c t_i)$, $t_i \sim t_5$ rescaled to $\text{Var}(t_i) = 1$ with $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma_c)))$. Panel a shows that the fit is slightly worse than in Table 6. Panel b shows that the distributional assumption does not meaningfully affect the qualitative results from the counterfactual simulations.

Table B11: Heterogeneity in energy savings

Baseline characteristic	n	Share = 1	Effect of adoption on total energy spending (US$/month): Without the characteristic	With the characteristic	Difference	p
Age above median	2081	0.48	-8.40	-9.93	-1.52	0.473
			(1.65)	(1.35)	(2.12)	
Charcoal spending above median	2081	0.49	-9.96	-8.30	1.65	0.439
			(1.45)	(1.56)	(2.13)	
Owns 2+ charcoal stoves	2081	0.07	-9.48	-9.72	-0.25	0.957
			(1.09)	(4.44)	(4.57)	
Owns at least one LPG stove	2081	0.57	-11.57	-7.84	3.73*	0.095
			(1.82)	(1.30)	(2.23)	
Ever cooks with firewood	2081	0.14	-8.98	-9.76	-0.78	0.768
			(1.18)	(2.37)	(2.65)	
Electricity spend above median	2081	0.30	-9.23	-9.12	0.12	0.958
			(1.32)	(1.77)	(2.21)	

Notes: Each row is a separate instrumental-variables regression of total monthly energy spending (charcoal, LPG, firewood and electricity, the last inclusive of the experimental subsidy) on adoption of the induction stove, adoption interacted with the binary baseline characteristic named in the row, the characteristic itself, willingness-to-pay, and willingness-to-pay interacted with the characteristic. Adoption and its interaction are both instrumented, by the randomly assigned BDM price, the credit treatment and their interaction, and by those three interacted with the characteristic. Controls are the stratification variables and the remaining randomized arms; adding the baseline covariate block of Table 1 leaves every conclusion intact. Pooled, the effect is -9.24 (s.e. 1.06) against a complier control mean of US$24.89 per month, with first-stage F statistics between 105 and 122. Age is split at the sample median of 35, monthly charcoal spending at US$16.17 and baseline electricity spending at US$3.08 per month; electricity spending is bunched at its median, which is why the high group is 30 rather than 50 percent of households. The charcoal-stove and firewood subgroups are small, at 7 and 14 percent of households. The LPG difference is the one estimate sensitive to the control set, falling to US$3.03 ($p = 0.17$) with the full block. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IV Additional background information

IV.1 Policy background

Clean cooking policy in East Africa Given steady population growth, the IEA projects that approximately one billion people in Africa will still cook with biomass as their primary cooking fuel in 2050 (IEA, 2025). The rapid rise of household electricity access combined with the persistent use of biomass fuel for cooking has generated widespread policy interest in 'leapfrogging' populations towards electric cookstoves. Per the Organization of the Petroleum Exporting Countries (OPEC) Fund for International Development, "most specialists agree that electric cooking from renewable energy sources remains the ideal solution over the longer term" (OPEC, 2024).

To mitigate the negative externalities associated with charcoal production and to stimulate revenue for the financially distressed electric utility, in 2017 Kenya's national electricity distributor launched "Pika na Power," or "cook with electricity" (Kenya Power, 2017). It aims to increase uptake of electric cooking "to 5% (500,000 customers) in the short term and to 10% in the medium term" (Kenya Power, 2023). The Kenyan government's "National Electric Cooking Strategy Action Plan" (Ministry of Energy, 2024b) announced VAT waivers for new electric appliances, behavior change campaigns, expanded support for consumer financing, investments in grid capacity and reliability, and an 'e-cooking' electricity tariff designed to make electric cooking more attractive relative to LPG and biomass stoves. This push follows Kenya's national electrification program, which increased household grid access to over 58% in 2024 (KNBS, 2024).

Other countries are following suit: Uganda's 2022 Nationally Determined Contribution (NDC) was for "electricity to reach 50% of cooking fuel share by 2025" (Republic of Uganda, 2022).

Standards and bans Standards and bans are common alternatives to the direct pricing of externalities (Fowlie et al., 2020; Goulder and Parry, 2008), and can be more efficient than taxes when customers misperceive energy costs (Houde and Myers, 2025). Kenya has promoted the adoption of energy-efficient lighting through CFL distribution programs and standards (Kovacova, 2026; Wambui, 2010), and is now one of Africa's largest LED markets (IMARC Group, 2025).

In the context of cooking, such policies face political and enforcement hurdles. Much of Kenya's charcoal-related deforestation is informal, and national logging bans have been largely ineffective due to enforcement gaps (Sola and Cerutti, 2021; Daghar, 2021). Traditional cookstoves are often produced locally and thus suffer from similar enforcement constraints. Given widespread informality in the charcoal and stove sectors, bans or standards likely require inefficiently costly enforcement.

IV.2 Credit policy

Customers can choose whether to make monthly, weekly or daily payments to help ease repayment. In the lead-up to each payment deadline, participants who have not yet paid receive SMS reminders. Delinquent accounts are manually checked and followed up on by a dedicated agent team (BURN Manufacturing, 2023). If the customer does not make the required payment within seven days, the device is remotely turned off and only turned back on once the owner makes the payment (a 'digital collateral' practice that is standard in East Africa; see Gertler et al., 2024). If after 14 days the owner has still not made the required payment, the agent visits the respondent in the field to either restructure the loan (if non-payment is due to unforeseen circumstances) or, if needed, to repossess the device.

IV.3 Charcoal stove temperature sensors

Charcoal stove usage During the baseline visit, enumerators install Stove Use Monitors (SUMs) to monitor usage of households' charcoal stoves. We use the EXACT Pro sensor, a thermistor-based logger measuring 56×26×12 mm, weighing 20 g, sealed against water and dust, and rated for continuous operation up to 120°C (Climate Solutions, 2024). The device takes a measurement every minute and writes to internal memory at a user-configured interval, which we set to two minutes; its capacity of 130,000 logs corresponds to roughly six months at this interval, and its battery lasts in excess of four years. The sensor measures the temperature of its stove-facing surface relative to ambient, so that the recorded series reflects heat from the stove rather than ambient warming. Data are downloaded once per month in the field using a handheld launcher that communicates with each sensor over Bluetooth Low Energy and writes to a microSD card. For households who owned two or more charcoal stoves at baseline, we installed monitoring devices on all stoves that were less than three years old (stoves older than this often caused the device to malfunction). The temperature sensors are drilled into the stove and securely fastened with an M4 self-tapping sheet-metal screw without damaging the functionality of the stove, positioned at the back or the outside of the stove body so as not to obstruct the cook; placement is held consistent across households so that the temperature series remain comparable. Figure 3 shows a charcoal stove instrumented with a temperature sensor. The installation occurs during the baseline visit such that we are able to estimate a within-individual treatment effect. Prior research has found minimal Hawthorne effects from electronic stove monitoring;[23] still, any Hawthorne effect resulting from the installation of stove sensors would be symmetric across treatment groups.

We develop a basic algorithm to convert the 2-minute temperature series to a 2-minute cooking dataset, which we then aggregate to the hourly level for the analyses (Figure B1 shows several example household-days of data). We restrict attention to complete household-days, defined as those with all 720 two-minute observations spanning a full 24 hours. For each household-day we construct two reference points from the day's own temperature distribution: a cold baseline, taken as the mean of the 15 coldest observations recorded between midnight and 5am, and a hot reference, taken as the mean of the five hottest observations of the day. Using these, an observation is marked as 'cooking' if either of two conditions holds: the temperature exceeds 40°C, or the temperature rose by at least 1°C in each of the three consecutive two-minute intervals ending at that observation. We then override this classification back to 'not cooking' when the stove is unambiguously cooling, defined as the temperature having fallen below the midpoint between that day's cold baseline and hot reference while failing to rise by 1°C in any of the three preceding intervals. The first condition captures sustained high heat, the second captures the sharp temperature rise at the start of a cooking event, and the override prevents the long cooling tail of a charcoal fire from being counted as continued use.

For each household we construct a complete two-minute grid running from the baseline visit through the data download visit, so that periods without sensor coverage remain explicit rather than being silently dropped. Two further adjustments then draw on the survey data. First, for households reporting at the endline survey that they no longer own a charcoal stove, all observations between the preceding visit and that visit are set to non-cooking. Second, we construct an imputed version of the series in which gaps in sensor coverage are set to non-cooking whenever the household reported zero charcoal expenditure over the corresponding recall period; we retain both the imputed and unimputed series. The two-minute indicators are then averaged within household-hour to give

[23] For example, in a randomized trial of improved cookstoves in Rwanda, Thomas et al. (2016) randomly assigned respondents to have a monitor installed in either a visible or hidden part of the cookstove. The visible monitor increases usage by 6% in the first week but causes no difference in cooking behavior from week two onwards.

the fraction of each hour spent cooking, which we scale to obtain hours and minutes of cooking per day, and which we split into the pre-period (between the baseline and the main visit) and the post-period (all observations thereafter).

IV.4 Willingness-to-pay

We measure WTP using the Becker et al. (1964) mechanism, building on the implementations developed in Berry et al. (2020), Dean (2024), and Berkouwer and Dean (2022). Neither the respondent nor the enumerator know the respondent's randomly assigned price. The enumerator conducts a binary search over the range of Ksh 1,500 to Ksh 18,000 (US$12 to US$139), first asking *"If the price of the ECOA is Ksh 9,792 [US$76] would you want to buy it?"*, then proceeding to a higher or lower price based on the respondent's answer, and repeating this process until arriving at the nearest Ksh 10 (US$0.08). After arriving at a final WTP, the envelope is opened and the respondent buys the stove for the price in the envelope if and only if it is less than or equal to their WTP. Since the price is fixed ex ante, the mechanism is incentive compatible. Since upfront prices are randomly assigned, purchasing is random conditional on WTP within the support of random prices. Practice BDM and take-it-or-leave-it (TIOLI) exercises result in similar estimates (Figure B5).

IV.5 Marginal-cost subsidy implementation

All respondents have pre-paid electricity meters, which is the norm in Kenya and across many African utilities. 87% of all Kenyan residential electricity customers have pre-paid meters, and 95% of all residential customers who were connected since 2015—which includes most low-income connected households—do. Consuming electricity runs down the meter: a customer must periodically 'top up' their meter to avoid it reaching zero (at which point electricity gets shut off).[24] They can do so by purchasing a unique 20-digit token code either at a kiosk or via the Kenya Power mobile application, and then manually entering this code on their meter. We implement electricity transfers through this same channel: the research team collects respondents' account numbers to purchase electricity tokens on their behalf, and then sends customers the 20-digit token code via SMS, which they then enter manually into their meter. The code is unique to the meter and cannot be transferred, sold, or returned. We process electric stove usage data, calculate subsidy amounts, and send respondents token codes on a weekly basis. For comparison, the median respondent in our sample purchases electricity four times per month, or once every 7.5 days, suggesting our implementation matches respondents' own purchasing habits.

On the one hand, the manual entry of tokens could add an additional hurdle that lowers the value of marginal-cost subsidies. On the other hand, it could make the transfer more salient than an automatic bill rebate. Rather than trying to bound these opposing forces, we only note that the utility recently used this same method of token transfers to implement a subsidy pilot study.[25] Our implementation methodology thus reflects the realistic implementation of such a program by the electric utility and the empirical estimates can be interpreted as policy relevant.

IV.6 Charcoal to CO_2e conversion

Intuitively, the mass of oxygen is approximately 16 g/mol and the mass of carbon (which charcoal largely consists of) is approximately 12 g/mol. Each carbon atom combines with two oxygen atoms

[24] For 31% of our sample this had happened once or twice in the past month. For the remainder it had not.

[25] Kenya Power's e-cooking tariff development study states that "Rebates are provided as electricity tokens sent via SMS to pre-paid customers" (Ministry of Energy, 2024a).

to generate CO_2 such that 12 grams of charcoal generates 44 grams of CO_2, or 3.7 kilograms CO_2 per kilogram of charcoal. This stoichiometric calculation is a useful benchmark, and it is close to the factor we would obtain if all of the biomass were harvested renewably and no emissions arose outside of combustion. However, it does not factor in production, processing, and transportation, which account for the majority of the total emissions factor; the purity of charcoal in East Africa (it does not consist exclusively of carbon); nor reforestation rates, which determine what share of combustion emissions is offset by regrowth.

To obtain a more accurate estimate of emissions in our context, we convert quantities of charcoal into CO_2e using an emissions factor that combines combustion and non-combustion sources. Combustion emissions depend on how much of the wood used to produce the charcoal is harvested non-renewably, since wood that regrows reabsorbs the carbon released when it is burned. Estimates from Floess et al. (2023) imply combustion emissions of 86.8 tCO_2e per terajoule for the non-renewable fraction of biomass and 7.8 tCO_2e per terajoule for the renewable fraction. We assume a fraction of non-renewable biomass (fNRB) of 38%, following the most recent MoFuSS estimates for Kenya (Ghilardi and Bailis, 2024; UNFCCC, 2023), which yields combustion emissions of 37.9 tCO_2e per terajoule. The production, processing, and transportation of charcoal add a further 124.2 tCO_2e per terajoule of non-combustion emissions (Gill-Wiehl et al., 2024; Whitman and Lehmann, 2011), for a total emissions factor of 162 tCO_2e per terajoule. With a calorific value of 0.027 terajoules per ton of charcoal, this yields an estimated 4.4 kilograms of CO_2e per kilogram of charcoal.

The fNRB assumption is the most consequential input into this calculation, since it governs the weight placed on the two combustion factors. Holding all else fixed, an fNRB of 10% lowers the emissions factor to 3.8 kilograms of CO_2e per kilogram of charcoal, while an fNRB of 100% raises it to 5.7 kilograms.

V Pre-Analysis Plan analysis

A pre-analysis plan, available here, was filed with the AEA RCT Registry, ID: 16832 (Berkouwer and Dean, 2025). This section reports each outcome listed under *Primary hypotheses for the environmental economics of induction stove adoption* in the pre-analysis plan, and points to where it appears in the paper. All estimates use the instrumental variables strategy the plan specified, instrumenting adoption with the randomly assigned price and credit treatments.

V.1 Preliminary outcomes

Minutes spent cooking with charcoal per day Charcoal cooking falls by 31 minutes per day, a 34% reduction against the complier control mean. Reported in column 1 of panel a of Table 1 and discussed in Section 4.1. Figure 4 shows the daily event study from 45 days before to 90 days after the main visit; Table A2 reports the specification alongside alternatives at the hourly and daily level, with and without respondent fixed effects.

Electricity expenditures and corresponding CO_2e emissions Monthly electricity spending rises by US$2.3 (65%), in column 5 of panel a of Table 1; column 6 reports the same among the subsidy control group. The corresponding emissions increase is in column 3 of panel b. Discussed in Section 4.2 and Section 4.3.

Charcoal expenditures and corresponding CO_2e emissions Monthly charcoal spending falls by US$10 (66%), in column 2 of panel a of Table 1. The corresponding emissions reduction is in column 1 of panel b.

LPG expenditures and corresponding CO_2e emissions Reported in column 3 of panel a of Table 1 and column 2 of panel b. The point estimates are small and not statistically distinguishable from zero.

Aggregate energy expenditures and corresponding CO_2e emissions Total monthly energy spending falls by US$8.9 (36%), in column 7 of panel a of Table 1, which is US$209 discounted over the stove's 2.2 year expected life. Total emissions fall by 2.5 tCO_2e per year, in column 4 of panel b, or 5.5 tCO_2e over the stove's life.

V.2 Primary outcomes: cost per ton of CO_2e abated

Fixed-cost subsidies, credit control group US$11 per tCO_2e. Reported in Section 5.1 and Table 3, with the sensitivity of the estimate to alternative assumptions in Table A3.

Fixed-cost subsidies, credit treatment group US$19 per tCO_2e, in the same table. The gap between the two is the paper's central result and is decomposed into its two channels in Table 3.

Electricity subsidies, credit treatment and credit control groups The pre-analysis plan anticipated a cost per ton for each group. We do not report one, because the marginal-cost subsidies moved neither margin: the effect on willingness-to-pay is a precise zero, and we can rule out an effect larger than US$0.2 per US$1 of subsidy (column 3 of Table A6); the usage elasticity is -0.03, ruling out a decrease larger than 0.12% in response to a 1% price increase (column 6). A cost per ton divides subsidy expenditure by the abatement that expenditure induced. Here the denominator is indistinguishable from zero, so the ratio is undefined rather than large, and the informative statement is the bound on each margin rather than a ratio of one bound to another. We therefore report the two margins separately, in Section 5.2 and Figure 5, with the SMS follow-up experiment in Figure A7 and Table B5.

This is a property of the results, not of the implementation. The subsidies were delivered as designed, through the same token channel households use to buy electricity and at the same weekly frequency, and take-up of the transfers was not the constraint; the SMS experiment was run precisely to test whether inattention rather than preferences explained the null, and found no effect either. The quantity the plan asked for is undefined because households did not respond to the price of electricity, which is itself one of the paper's findings. For the same reason, we are also unable to separate the selection and treatment channels in the effect of the announced subsidy on emissions.

V.3 Determinants of aggregate subsidy efficiency

Marginal adopters incentivized per subsidy dollar Reported as its inverse, the subsidy expenditure required per additional adoption, which includes payments to inframarginal adopters: US$65 in the credit control group and US$93 in the credit treatment group. In Table 3, discussed in Section 5.1. The underlying demand curves are in Figure 5 and the elasticities in Table 5.

Externality avoided by each marginal adopter 2.6 tCO_2e per year in the credit control group and 2.2 tCO_2e in the credit treatment group, or 5.7 and 4.8 over the stove's life. Reported in Table 4 and Table 3. The four proxies the plan named are reported separately: charcoal stove usage in Figure 4 and column 1 of Table 1, induction stove usage in panel b of Figure 5, and charcoal and electricity emissions in columns 1 and 3 of panel b of Table 1.

V.4 Secondary outcomes

Time savings None detected. Charcoal cooking falls by 27 minutes per day while induction cooking adds 37 minutes, leaving households cooking 8.2 minutes more in total, with both stoves in use simultaneously only 0.7% of stove-time. Reported in columns 7–9 of Table A2 and discussed in Section 4.4.

Self-reported health Reductions in self-reported cough and breathlessness among respondents and their children. Reported in Berkouwer and Dean (2026) rather than here, as part of that paper's treatment of the household-level impacts of adoption.

Whether WTP and induction stove usage vary by underlying power quality Reported in Berkouwer and Dean (2026), which measures outages and voltage at the customer's own outlet and treats grid reliability as a moderator of adoption and usage.